

Group Theory and Applications

Lecture Notes

Semester I, 2025-26

Course taught by Prof. Bhavesh Chauhan

Notes by Jay Agarwal

Contents

Lecture 1: Preliminaries	5
1.1 Introduction	5
1.2 Algebraic Structures	6
1.3 Mappings	6
Lecture 2: Groups	8
2.1 Groups	8
2.2 Examples of Groups	9
2.3 Subgroups	10
Lecture 3: Cyclic and Dihedral Groups	11
3.1 Order and Cayley Tables	11
3.2 Cyclic Groups	11
3.3 Dihedral Groups	12
Lecture 4: Generators, Subgroups and Cosets	14
4.1 Generators	14
4.2 Subgroups	14
4.3 Cosets	15
Lecture 5: The Symmetric Group and Isomorphism	17
5.1 The Symmetric Group	17
5.2 Isomorphism	18
Lecture 6: Normal Subgroups	20
6.1 Two ways to break up a group	20
6.2 Normal subgroups	20

6.3	Cosets partition the group	21
Lecture 7: Quotient Groups		22
7.1	Cosets are identical or disjoint	22
7.2	Multiplying cosets	23
7.3	Quotient groups	23
Lecture 8: The Coset Product and Homomorphisms		24
8.1	Products of sets	24
8.2	The coset product	24
8.3	D_3/H_4 is a group	25
8.4	Homomorphisms	25
8.5	Conjugate elements	26
Supplement A: Conjugacy Classes		27
A.1	Conjugation is an equivalence relation	27
A.2	Conjugacy classes	27
A.3	The centralizer and the class equation	28
A.4	The centre	28
A.5	Conjugate subgroups	29
A.6	Cauchy's and Sylow's theorems	29
Lecture 9: Direct Products and Kernels		30
9.1	External direct product	30
9.2	Internal direct product	30
9.3	The kernel of a homomorphism	32
9.4	Normal subgroups and conjugacy classes	32

Lecture 10: Representations	33
10.1 Representations	33
10.2 Group actions	34
10.3 A two-dimensional representation of D_3	34
10.4 Equivalent representations	36
10.5 Reducible representations	36
10.6 Characters	37
 Lecture 11: Maschke's Theorem and Vector Spaces	 38
11.1 Similarity transformations	38
11.2 Maschke's theorem	39
11.3 Vector spaces	39
11.4 Scalar products	40
 Lecture 12: Unitary Representations and Schur's Lemma	 41
12.1 Representations of a quotient group	41
12.2 Invariant subspaces	41
12.3 Orthonormal bases	42
12.4 Unitary representations	42
12.5 Schur's lemma	43
12.6 The character table of D_3	43
 Lecture 13: Orthogonality and Character Tables	 44
13.1 The character table of C_2	44
13.2 The great orthogonality theorem	44
13.3 Orthogonality of characters	45
13.4 The character table of C_3	45

13.5	The character table of D_3	46
Lecture 14: Schur's Second Lemma and Decomposition		48
14.1	Schur's second lemma	48
14.2	Orthogonality of matrix elements	48
14.3	Decomposing a representation	49
14.4	Three-dimensional representations of D_3	50
Lecture 15: Direct Products of Representations		51
15.1	The direct product of two representations	51
15.2	Reducing the direct product	51
15.3	Example: $2 \otimes 2$ in D_3	51
15.4	Orthogonality and completeness	52
15.5	A test for irreducibility	52
Lecture 16: Worked Example: D_4		53
16.1	The problem	53
16.2	The group	53
16.3	Conjugacy classes	53
16.4	How many representations, and of what dimension	54
16.5	The one-dimensional representations	54
16.6	The two-dimensional representation	55
16.7	The character table	55
16.8	Reducing $2 \otimes 2$	56

Lecture 1

Preliminaries

1.1 Introduction

These notes were taken during the course *Group Theory and Applications*, taught in Semester I of 2025–26 by Prof. Bhavesh Chauhan.

The course divides into three parts.

The **first part** sets up the basic definitions and key concepts of group theory, working largely with finite groups. The main sources here are Chapter 2 of Herstein and Chapter 2 of Wu-Ki Tung.

The **second part** develops representation theory, with the emphasis on irreducible representations. This follows Chapter 3 of Wu-Ki Tung together with Chapters 3 and 4 of Jones.

The **third part** turns to Lie groups and Lie algebras, where Georgi becomes the primary text for the remainder of the course. Zee is used for particular topics, and Wu-Ki Tung and Jones serve as supplements throughout.

The course closes with a discussion of spontaneous symmetry breaking and a big-picture view of particle physics

A note on how to use these notes.

These notes are not rigorous and are not trying to be. A good deal of what follows is hand-waved or simplified past the point a mathematician would accept. That reflects the course itself: it is an introduction and the lectures were pitched at a level everyone could follow.

The format was that lectures introduced the essentials without going too deep, students read the details on their own afterwards, and questions were brought back to the following class. These notes correspond to the first of those three steps only.

So read them as a guide to what the course covers and in what order, not as a self-contained treatment. The proofs, the worked examples, and the details that have been compressed or skipped here are all in the texts, done properly. These notes assume you are reading the texts alongside them and they will be of limited use otherwise.

On accuracy. These notes were digitized from my handwritten notes using AI, then verified and edited by me. Mistakes may still have slipped through. If you spot one, please let me know.

Text Books

- I. N. Herstein, *Topics in Algebra*.
- Wu-Ki Tung, *Group Theory in Physics*.
- H F Jones, *Groups, Representations and Physics*
- A. Zee, *Group Theory in a Nutshell for Physicists*.
- H. Georgi. *Lie Algebras in Particle Physics*

1.2 Algebraic Structures

An algebraic structure is a set together with one or more operations on it, subject to axioms. A set is simply a collection of objects, without any operation. The structures below are listed roughly in order of how much they demand. Please ignore any terms in these bullets which you don't know.

- **Group** - a set with one binary operation, we'll talk more about it in the next lecture
- **Ring** - a set with two operations, refer to Herstein Chapter 3 for more details
- **Field** - a ring in which every non-zero element has a multiplicative inverse. *Examples:* $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$. We are only concerned with $\mathbb{R}$, $\mathbb{C}$ for the course.
- **Vector space** - an abelian group V together with a field F of scalars, closed under both operations:

$$v \oplus w \in V, \quad \alpha v \in V, \quad \forall v, w \in V, \alpha \in F.$$

More on this later, when we study representation theory

1.3 Mappings

Before we study Groups, let's briefly discuss mappings.

Mapping. If S and T are non-empty sets, a mapping from S to T is a subset $M \subseteq S \times T$ such that for every $s \in S$ there is a unique $t \in T$ with $(s, t) \in M$. We write $\sigma : S \rightarrow T$ for such a mapping, and $\sigma(s) = t$, or sometimes $s\sigma = t$.

Order of composition. We write mappings on the right of their arguments and compose left to right. So for $\sigma : S \rightarrow T$ and $\tau : T \rightarrow U$,

$$s(\sigma \circ \tau) := (s\sigma)\tau,$$

that is, σ acts first.

This is Herstein's convention, and we use it in this lecture. Much of the literature uses the opposite one, in which $f \circ g$ means " g first". These notes follow whichever convention the source being used follows. The mathematics is identical either way, but products are reversed, so it is always worth checking which convention is in use.

The set A_S . For a set S , let

$$A_S = \{ \sigma \mid \sigma : S \rightarrow S \text{ is one-to-one and onto} \}$$

be the set of all bijections of S onto itself.

Properties of A_S . Composition of bijections satisfies the following. Try to reason why.

Closure

If $\sigma_1, \sigma_2 \in A_S$ then $\sigma_1 \circ \sigma_2 \in A_S$.

Associativity

If $\sigma_1, \sigma_2, \sigma_3 \in A_S$ then $\sigma_1 \circ (\sigma_2 \circ \sigma_3) = (\sigma_1 \circ \sigma_2) \circ \sigma_3$.

Identity

$\exists e \in A_S$ such that $\sigma \circ e = e \circ \sigma = \sigma$ for all $\sigma \in A_S$.

Inverse

For every $\sigma \in A_S$ there is $\sigma^{-1} \in A_S$ with $\sigma^{-1} \circ \sigma = \sigma \circ \sigma^{-1} = e$.

Can you see which of these properties is the one that really needs bijectivity? Hint: it starts with I.

Composition need not commute.

Exercise

For $\sigma, \tau \in A_S$, is $\sigma \circ \tau = \tau \circ \sigma$?

Solution: No, three elements already suffice. Let $S = \{x_1, x_2, x_3\}$ and define $\sigma, \tau : S \rightarrow S$ by

$$\begin{array}{ll} x_1\sigma = x_2 & x_1\tau = x_1 \\ x_2\sigma = x_3 & x_2\tau = x_3 \\ x_3\sigma = x_1 & x_3\tau = x_2 \end{array}$$

so σ cycles all three elements and τ swaps the last two.

Composing with σ first:

$$\begin{aligned} x_1(\sigma \circ \tau) &= (x_1\sigma)\tau = x_2\tau = x_3, \\ x_2(\sigma \circ \tau) &= (x_2\sigma)\tau = x_3\tau = x_2, \\ x_3(\sigma \circ \tau) &= (x_3\sigma)\tau = x_1\tau = x_1. \end{aligned}$$

Composing with τ first:

$$\begin{aligned} x_1(\tau \circ \sigma) &= (x_1\tau)\sigma = x_1\sigma = x_2, \\ x_2(\tau \circ \sigma) &= (x_2\tau)\sigma = x_3\sigma = x_1, \\ x_3(\tau \circ \sigma) &= (x_3\tau)\sigma = x_2\sigma = x_3. \end{aligned}$$

The two disagree already on x_1 , since $x_1(\sigma \circ \tau) = x_3$ while $x_1(\tau \circ \sigma) = x_2$. Hence

$$\boxed{\sigma \circ \tau \neq \tau \circ \sigma}$$

Lecture 2

Groups

2.1 Groups

Group: A non-empty set of elements G is said to form a *group* if there exists a binary operation $\circ$ on G , called the *product* or *group multiplication*, such that:

(1) If $g_\alpha, g_\beta \in G$, then $g_\alpha \circ g_\beta \in G$. (closure)

(2) If $g_\alpha, g_\beta, g_\gamma \in G$, then

$$g_\alpha \circ (g_\beta \circ g_\gamma) = (g_\alpha \circ g_\beta) \circ g_\gamma.$$

(associativity)

(3) There exists $g_0 \in G$ such that for all $g_\alpha \in G$,

$$g_0 \circ g_\alpha = g_\alpha \circ g_0 = g_\alpha.$$

(identity, or neutral element)

(4) For all $g_\alpha \in G$ there exists $g_\alpha^{-1} \in G$ such that

$$g_\alpha^{-1} \circ g_\alpha = g_\alpha \circ g_\alpha^{-1} = g_0.$$

(inverse)

These are exactly the four properties of A_S from the previous lecture. The definition above is what remains once the mappings are forgotten and only their behaviour under composition is kept.

Homework

A *semi-group* $(S, \circ)$ is a weaker structure than a group, required to satisfy only closure and associativity, with no identity and no inverses. Look into what survives without those two axioms, and find examples of semi-groups that are not groups.

Exercise

Find a set and an operation for which the associative law fails.

Abelian group: A group in which $g_\alpha \circ g_\beta = g_\beta \circ g_\alpha$ for all elements is called *abelian* or *commutative*. The integers under addition and the non-zero reals under multiplication are abelian; A_S from the last lecture is not.

Weakening the axioms: Axioms (3) and (4) each demand something on both sides. It turns out this is more than necessary: if we assume only a *left* identity and a *left* inverse, the right-handed versions follow.

Suppose $g_0 \circ g_\alpha = g_\alpha$ and $g_\alpha^{-1} \circ g_\alpha = g_0$ for every $g_\alpha \in G$. Set

$$x = g_\alpha \circ g_\alpha^{-1}.$$

Then

$$x \circ x = g_\alpha \circ \underbrace{(g_\alpha^{-1} \circ g_\alpha)}_{g_0} \circ g_\alpha^{-1} = g_\alpha \circ \underbrace{(g_0 \circ g_\alpha^{-1})}_{g_\alpha^{-1}} = g_\alpha \circ g_\alpha^{-1} = x.$$

Now x has a left inverse x^{-1} by assumption, so multiplying $x \circ x = x$ on the left by x^{-1} gives

$$\underbrace{(x^{-1} \circ x)}_{g_0} \circ x = x^{-1} \circ x = g_0 \implies x = g_0.$$

That is,

$$\boxed{g_\alpha \circ g_\alpha^{-1} = g_0}$$

so the right inverse exists as well. A similar argument recovers the right identity.

2.2 Examples of Groups

Finite groups under multiplication. Each of the following is a group under ordinary multiplication of complex numbers:

- $\{1\}$, the trivial group.
- $\{-1, 1\} \cong \mathbb{Z}_2$.
- $\{1, i, -1, -i\} \cong \mathbb{Z}_4$, the fourth roots of unity.
- $\{1, \omega, \omega^2\}$, the cube roots of unity.

All of these are abelian, since ordinary multiplication is.

The symbol $\cong$ means the two groups are *isomorphic*: there is a one-to-one correspondence between their elements that respects the group operation, so they have identical structure even though the elements themselves may be quite different objects. We will make this precise later.

The two-element group. There is essentially only one group with two elements. Write it as $\{e, a\}$. The only thing to decide is what $a \circ a$ should be. If $a \circ a = a$, then multiplying by a^{-1} on the right gives

$$a \circ a \circ a^{-1} = a \circ a^{-1} \implies a = e,$$

contradicting $a \neq e$. So $a \circ a = e$, and the multiplication table is forced:

$\circ$	e	a
e	e	a
a	a	e

N th roots of unity. Generalising the finite examples above,

$$\Omega_N = \{1, \omega, \omega^2, \dots, \omega^{N-1}\} = \left\{ e^{2\pi i j / N} \mid j = 0, 1, 2, \dots, N-1 \right\}$$

is a group under multiplication for every N .

The unit circle. Take

$$G = \{ c \in \mathbb{C} \mid |c| = 1 \}.$$

Closure holds because $|c_1 c_2| = |c_1| |c_2| = 1$, and the identity is 1. For inverses, if $b \circ c_1 = c_1 \circ b = 1$ then $|b| = 1$, so b is again in G . This makes G a group under ordinary multiplication.

Parametrising instead of checking element by element is often easier: every element of G can be written as $e^{i\phi}$, which gives the whole set at once. This group is called $U(1)$:

$$U(1) = \{ e^{i\phi} \mid \phi \in \mathbb{R} \}$$

Note that $\Omega_N \subset U(1)$ for every N .

Integers modulo N . The set $\mathbb{Z}_N = \{0, 1, 2, \dots, N-1\}$ under addition modulo N is a group. It is the same group as Ω_N written additively.

Rotations in two dimensions. $R_{2D}(\theta)$ needs one parameter, and $R(\theta_1) \circ R(\theta_2) = R(\theta_1 + \theta_2)$. Since $\theta_1 + \theta_2 = \theta_2 + \theta_1$, this is an abelian group.

Rotations in three dimensions. $R_{3D}(\theta, \phi, \psi)$ needs three parameters. Two rotations still compose to give a rotation, and associativity, identity and inverses hold, so this is again a group. But it is *non-abelian*: rotating a book about the vertical axis and then a horizontal one leaves it in a different orientation than doing the two in the opposite order.

2.3 Subgroups

Subgroup. A subset $H \subseteq G$ that itself forms a group under the same operation as G is called a *subgroup* of G .

For example, with $G = \{1, -1, i, -i\}$ under ordinary multiplication, the subset

$$H = \{1, -1\}$$

is a subgroup, since it is closed, contains the identity, and each element is its own inverse.

Lecture 3

Cyclic and Dihedral Groups

3.1 Order and Cayley Tables

Order: For a finite group, the *order* is the number of elements in the group.

Cayley table: A group multiplication table, listing the product of every pair of elements. Rows and columns are labelled by the group elements, and the entry in row a , column b is $a \circ b$.

Groups of order 2: Write the group as $\{e, g\}$. The only thing to decide is $g \circ g$, and there are two candidates:

$$g \circ g = g \quad \text{or} \quad g \circ g = e.$$

If $g \circ g = g$, multiply on the left by g^{-1} :

$$g^{-1} \circ (g \circ g) = g^{-1} \circ g \implies g = e,$$

which contradicts $g \neq e$. Otherwise we are just left with the trivial group $\{e\}$. So $g \circ g = e$ is forced, and the table is

$\circ$	e	g
e	e	g
g	g	e

There is only one group of order 2.

Groups of order 3: The same kind of reasoning fixes $\{e, g_1, g_2\}$ completely, with $g_1^{-1} = g_2$:

$\circ$	e	g_1	g_2
e	e	g_1	g_2
g_1	g_1	g_2	e
g_2	g_2	e	g_1

Every entry in a Cayley table is forced by the requirement that each element appear exactly once in each row and column.

Commutativity: A group is abelian precisely when its Cayley table is symmetric about the diagonal. We can clearly see that the above two groups are both abelian.

3.2 Cyclic Groups

Both tables above are examples of *cyclic groups*: groups generated by repeatedly applying a single element.

$$C_n = \{a^0, a^1, a^2, \dots, a^{n-1}\}, \quad a^n = a^0 = e.$$

The element a is called a *generator*. For $\{1, \omega, \omega^2\}$, the cube roots of unity, ω is a generator.

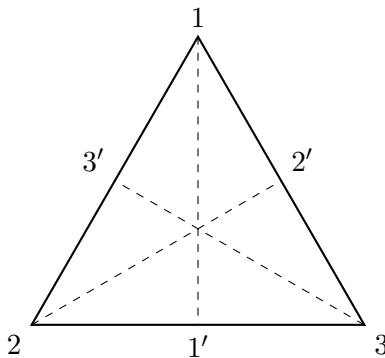
Cyclic groups are always abelian, since $a^i \circ a^j = a^{i+j} = a^j \circ a^i$.

Where they show up? C_2 describes parity and time reversal. C_3 describes the rotations of an equilateral triangle.

3.3 Dihedral Groups

Dihedral group D_n : The symmetries of a regular n -sided polygon, meaning all the ways it can be moved and end up looking the same. Its order is $2n$: n rotations and n reflections.

Example D_3 : Label the vertices of an equilateral triangle 1, 2, 3. Dropping a line from vertex 1 to the midpoint of the opposite side gives an axis $11'$, and flipping about it leaves the triangle looking unchanged. It swaps vertices 2 and 3, so as a permutation this reflection is (23) . The axes $22'$ and $33'$ give the other two reflections, (31) and (12) .



Rotating by 120° and 240° (that is, $2\pi/3$ and $4\pi/3$) gives the permutations (123) and (321) . Together with the identity:

$$D_3 = \{ e, (12), (23), (31), (123), (321) \}$$

which has order $6 = 2 \times 3$, as expected.

Note: The notation (123) means the cycle sending $1 \rightarrow 2$, $2 \rightarrow 3$ and $3 \rightarrow 1$, so (321) sends $3 \rightarrow 2$, $2 \rightarrow 1$ and $1 \rightarrow 3$. A two-element cycle like (23) just swaps the two. We will look at permutations properly later.

Exercise

Work out the full Cayley table for D_3 yourself, one entry at a time, tracking where each of 1, 2, 3 ends up. Remember that the right element acts first. Check two things when you are done: every element appears exactly once in each row and each column, and the table is not symmetric about the diagonal.

Cayley table for D_3 . Here $a \circ b$ means b acts first.

$\circ$	e	(12)	(23)	(31)	(123)	(321)
e	e	(12)	(23)	(31)	(123)	(321)
(12)	(12)	e	(123)	(321)	(23)	(31)
(23)	(23)	(321)	e	(123)	(31)	(12)
(31)	(31)	(123)	(321)	e	(12)	(23)
(123)	(123)	(31)	(12)	(23)	(321)	e
(321)	(321)	(23)	(31)	(12)	e	(123)

The table is not symmetric: $(1\ 2) \circ (2\ 3) = (1\ 2\ 3)$ while $(2\ 3) \circ (1\ 2) = (3\ 2\ 1)$. So D_3 is non-abelian, and it is the smallest non-abelian group.

Homework

Find all the symmetries of a rectangle (not a square) and list the group elements. What is the order of this group? Is it abelian? Compare it with D_3 and with the cyclic group of the same order.

Lecture 4

Generators, Subgroups and Cosets

4.1 Generators

Order of an element. If $G = \langle a \rangle$ with $a^n = e$, then $|a| = n$ and

$$G = \{e, a, a^2, \dots, a^{n-1}\}.$$

We also write $|G|$ for the order of the group, that is, the number of elements in it.

More than one generator. For $G = \langle a, b \rangle$, the group consists of all words that can be built from a and b :

$$G = \{g \mid g = a^{n_1} b^{m_1} a^{n_2} b^{m_2} \dots\}.$$

Take $G = \langle a, b \rangle$ with $a^2 = b^2 = e$. The relations let us cancel repeats, but words can still grow indefinitely:

$$G = \{e, a, b, ab, ba, aba, bab, \dots\}.$$

Adding the requirement that G be abelian, so $ab = ba$, collapses this to four elements:

$$G = \{e, a, b, ab\}.$$

This is the *Klein four-group*, written V_4 . Its Cayley table is

$\circ$	e	a	b	ab
e	e	a	b	ab
a	a	e	ab	b
b	b	ab	e	a
ab	ab	b	a	e

Every element is its own inverse, and the table is symmetric, so V_4 is abelian. It has order 4 but is not cyclic, since no single element generates it.

4.2 Subgroups

Subgroup: Given a group $(G, \circ)$, a subset $H \subseteq G$ that itself forms a group under $\circ$ is a *subgroup* of G . Both $\{e\}$ and G are always subgroups, called the trivial ones.

Take $G = \{e, a, b, ab\} = V_4$. The candidates of order 2 all work:

$$H_1 = \{e, a\}, \quad H_2 = \{e, b\}, \quad H_3 = \{e, ab\}.$$

The candidates of order 3 all fail:

$$\{e, a, b\}, \quad \{e, a, ab\}, \quad \{e, b, ab\}$$

are not closed. Can you think why?

Lagrange's theorem: If H is a subgroup of G , then $|H|$ divides $|G|$:

$$\frac{|G|}{|H|} \in \mathbb{Z}.$$

This is why the order-3 candidates above had no chance: 3 does not divide 4.

Rearrangement lemma: For $p, a, b \in G$, if $p \circ a = p \circ b$ then $a = b$. Consequently, multiplying every element of G by a fixed g just permutes them:

$$gG = G, \quad Gg = G.$$

Exercise

Let $|G| = p$ where p is prime. What are the possible subgroups of G ? What does this force G itself to be?

Exercise

Take $G = \{1, i, -1, -i\}$ under multiplication and fix $g = i$. Write out gG and check that you get all four elements back, just in a different order. Now do the same for the Cayley table of D_3 : pick any row and confirm that no element appears twice.

4.3 Cosets

Coset: Let H be a subgroup of G and $g \in G$. Then gH is a *left coset* and Hg is a *right coset* of H .

For V_4 with $H_1 = \{e, a\}$:

$$bH_1 = \{b, ab\}, \quad abH_1 = \{ab, b\}.$$

The cosets cover the whole group:

$$G = \bigcup_{g \in G} gH$$

For an abelian group, $gH = Hg$ always, since the elements commute. So in V_4 we have $H_1b = bH_1$.

Homework

For a non-abelian group the left and right cosets need not agree. Take D_3 and:

- find all its subgroups;
- find the subgroups H for which $gH = Hg$ for every $g \in D_3$.

Solution: Recall

$$D_3 = \{e, (12), (23), (31), (123), (321)\},$$

so $|D_3| = 6$ and by Lagrange any subgroup has order 1, 2, 3 or 6. The non-trivial ones are

$$H_1 = \{e, (12)\}, \quad H_2 = \{e, (23)\}, \quad H_3 = \{e, (31)\}, \quad H_4 = \{e, (123), (321)\}.$$

Sets like $\{e, (12), (31)\}$ fail on closure, and $\{e, (123)\}$ fails because $(123) \circ (123) = (321)$ is not in it. The rotations do close among themselves:

$\circ$	e	(123)	(321)
e	e	(123)	(321)
(123)	(123)	(321)	e
(321)	(321)	e	(123)

Now check the cosets. For H_1 , taking $g = (23)$:

$$(23)H_1 = \{(23), (321)\}, \quad H_1(23) = \{(23), (123)\},$$

which differ. The same happens for H_2 and H_3 .

For H_4 every element works. For instance

$$(12)H_4 = \{(12), (23), (31)\}, \quad H_4(12) = \{(12), (31), (23)\},$$

which are the same set. Checking the remaining elements the same way,

$$gH_4 = H_4g \quad \text{for all } g \in D_3.$$

So H_4 is the only non-trivial subgroup with this property. Subgroups that commute with every element in this way are called *normal*, and they will matter a great deal later.

Lecture 5

The Symmetric Group and Isomorphism

5.1 The Symmetric Group

Symmetric group S_n : The group of all permutations of n objects, with $|S_n| = n!$.

Two-row notation: A permutation $p \in S_n$ is written by listing where each object goes:

$$p = \begin{pmatrix} 1 & 2 & 3 & \cdots & n \\ p(1) & p(2) & p(3) & \cdots & p(n) \end{pmatrix}$$

The top row is the label, the bottom row the image.

Composition: For $p, q \in S_n$,

$$p \circ q = \begin{pmatrix} 1 & 2 & 3 & \cdots & n \\ p(q(1)) & p(q(2)) & p(q(3)) & \cdots & p(q(n)) \end{pmatrix}$$

so q acts first, matching the convention used from Lecture 3 onwards.

For example, in S_3 take

$$p = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}, \quad q = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}.$$

Then q sends $1 \rightarrow 3$ and p fixes 3, so $1 \rightarrow 3$. Similarly $2 \rightarrow 2 \rightarrow 1$ and $3 \rightarrow 1 \rightarrow 2$, giving

$$p \circ q = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}.$$

Homework

Show that S_n under composition is a group.

Elements of S_3 : There are $3! = 6$ of them. The identity and the three two-cycles are

$$e = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, \quad (12) = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}, \quad (23) = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}, \quad (31) = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

and the two three-cycles are

$$(123) = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, \quad (321) = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}.$$

So

$$S_3 = \{e, (12), (23), (31), (123), (321)\}.$$

This is exactly the same as D_3 ! What's happening?

Multiplying in cycle notation: Composition works the same way, but you can read it straight off the cycles. Take $(12) \circ (123)$, with the right one acting first:

$$1 \xrightarrow{(123)} 2 \xrightarrow{(12)} 1, \quad 2 \xrightarrow{(123)} 3 \xrightarrow{(12)} 3, \quad 3 \xrightarrow{(123)} 1 \xrightarrow{(12)} 2.$$

So 1 is fixed while 2 and 3 swap:

$$(12) \circ (123) = (23).$$

In two-row notation the same calculation reads

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}.$$

Warning: the bottom row is not cycle notation. Take $\begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} = (23)$. The bottom row reads 1 3 2, but the permutation is the swap (23), not the cycle (132). The bottom row lists images position by position; a cycle lists a chain of elements each mapping to the next.

Cycle notation in general: Read off cycles by following each object until it returns to where it started. For example, in S_6 ,

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 5 & 4 & 1 & 2 & 6 \end{pmatrix} = (134)(25)(6).$$

Here $1 \rightarrow 3 \rightarrow 4 \rightarrow 1$ closes the first cycle, $2 \rightarrow 5 \rightarrow 2$ the second, and 6 is fixed. Fixed points are usually left out, so this is often just written $(134)(25)$.

Exercise

Every permutation can be written as a product of two-cycles, for example $(123) = (12) \circ (23)$. This decomposition is not unique, but the number of two-cycles is always either even for a given permutation or always odd. Check this for the elements of S_3 by finding two different decompositions of (123) , and see which elements of S_3 are even and which are odd.

5.2 Isomorphism

Isomorphism: Two groups G and G' are *isomorphic*, written $G \cong G'$ or $G \leftrightarrow G'$, if there is a one-to-one correspondence $g \leftrightarrow g'$ between their elements, $g \in G$ and $g' \in G'$, that preserves the law of group composition. That is, whenever $g_i \leftrightarrow g'_i$ and $g_j \leftrightarrow g'_j$,

$$g_i \circ g_j = g_k \quad \implies \quad g'_i \circ' g'_j = g'_k,$$

where $\circ$ and $\circ'$ are the operations on G and G' .

The point is that isomorphic groups have the same structure. The elements may be permutations in one and complex numbers in the other, but the Cayley tables match once the labels are lined up.

Example: $D_3 \cong S_3$: the symmetries of the triangle are exactly the permutations of its three vertices. And

$$C_3 = \{e, a, a^2\} \cong \{e, (123), (321)\},$$

since $(123) \circ (123) = (321)$ plays the role of a^2 .

Exercise

Is $C_4 \cong V_4$? Both have order 4, so compare their Cayley tables and see whether the elements can be paired up.

Cayley's theorem: Every finite group of order n is isomorphic to a subgroup of S_n . The proof and further examples are in the books.

This is a very powerful statement, permutation groups are not one family among many, they are the whole story for finite groups. Any finite group can be realised as permutations of its own elements.

Lecture 6

Normal Subgroups

6.1 Two ways to break up a group

There are two ways to split a group into smaller pieces, and they turn out to be the same picture seen from two sides.

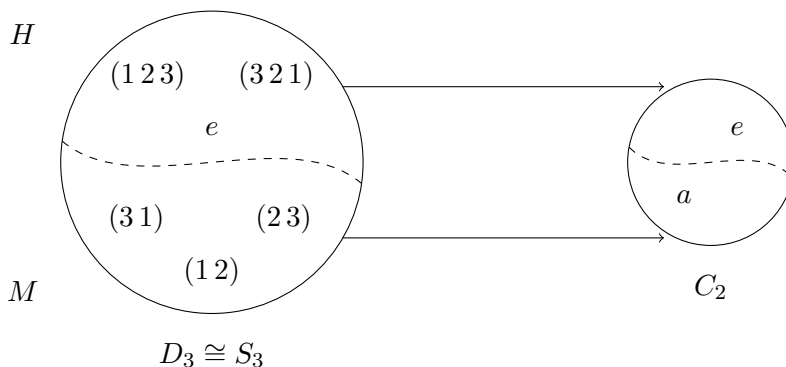
The first is by **cosets**, which we met in Lecture 4. Given a subgroup $H \subseteq G$, the group is covered by the cosets of H :

$$G = \bigcup_{g \in G} gH.$$

The second is by **homomorphisms**. A homomorphism is a map from one group to another that preserves the group operation: the product of two elements maps to the product of their images. We will define it properly later.

Take a map $\varphi : G \rightarrow G'$ onto a smaller group G' . The elements of G sent to the identity of G' form a subgroup H . Each of the remaining elements of G' comes from exactly one coset of H , so φ hands us the same partition of G that the cosets of H do.

For D_3 , take $H = \{e, (1\ 2\ 3), (3\ 2\ 1)\}$, the rotations, and let M be the remaining three elements, the reflections. Sending every rotation to e and every reflection to a lands us in the two-element group $\{e, a\}$ from Lecture 2.



6.2 Normal subgroups

Normal subgroup. A subgroup $H \subseteq G$ is *normal* or *invariant* if

$$gH = Hg \quad \text{for every } g \in G.$$

Equivalently, $gHg^{-1} = H$ for every $g \in G$. The two say the same thing, since multiplying $gH = Hg$ on the right by g^{-1} gives one from the other. The second form is often the easier one to check.

A warning about what this does not say. $gH = Hg$ is an equality of *sets*, not of elements. It does not require $gh = hg$ for each individual $h \in H$. What it does require is that the elements ghg^{-1} , as h runs over H , stay inside H and cover all of it.

In an abelian group every subgroup is normal, since $gh = hg$ holds element by element. The interesting cases are all non-abelian.

We saw both behaviours in Lecture 4. In D_3 the rotations $H_4 = \{e, (123), (321)\}$ satisfy $gH_4 = H_4g$ for all g , so H_4 is normal in D_3 . The order-2 subgroups H_1, H_2, H_3 are not.

6.3 Cosets partition the group

Every subgroup, normal or not, partitions the larger group into non-overlapping sets of equal size. Those sets are the cosets.

The partition can be built from left cosets aH or from right cosets Ha . For a normal subgroup the two partitions are identical, which is what $gH = Hg$ says. For a non-normal subgroup they are different.

Homework

In Lecture 4 we found that $H_1 = \{e, (12)\}$, $H_2 = \{e, (23)\}$ and $H_3 = \{e, (31)\}$ are the non-normal subgroups of D_3 . Work out how each of them partitions D_3 , into left cosets and into right cosets. In every case you should get D_3 as a disjoint union of cosets.

Solution for H_1 . Recall

$$D_3 = \{e, (12), (23), (31), (123), (321)\}, \quad H_1 = \{e, (12)\}.$$

Products follow the convention from Lecture 3, so the rightmost factor acts first.

Left cosets.

$$eH_1 = \{e, (12)\}, \quad (123)H_1 = \{(123), (31)\}, \quad (321)H_1 = \{(321), (23)\}.$$

The other three elements give nothing new. For instance $(12)H_1 = \{(12), e\}$, which is eH_1 again. So

$$D_3 = \underbrace{\{e, (12)\}}_{eH_1} \cup \underbrace{\{(123), (31)\}}_{(123)H_1} \cup \underbrace{\{(321), (23)\}}_{(321)H_1}.$$

Right cosets.

$$H_1e = \{e, (12)\}, \quad H_1(123) = \{(123), (23)\}, \quad H_1(321) = \{(321), (31)\}.$$

$$D_3 = \underbrace{\{e, (12)\}}_{H_1e} \cup \underbrace{\{(123), (23)\}}_{H_1(123)} \cup \underbrace{\{(321), (31)\}}_{H_1(321)}.$$

Both are genuine partitions: three blocks of two, covering all six elements with no overlap. But they are not the same partition. The block containing (123) is $\{(123), (31)\}$ on the left and $\{(123), (23)\}$ on the right. This is H_1 failing to be normal, seen as a statement about partitions rather than about individual products.

H_2 and H_3 work out the same way. Try them.

Lecture 7

Quotient Groups

7.1 Cosets are identical or disjoint

Exercise

Show that two left cosets of a subgroup H are either the same set or have no element in common. The same holds for right cosets.

Solution: Take two cosets pH and qH . We want

$$pH \cap qH = \emptyset \quad \text{or} \quad pH = qH.$$

Suppose they are not disjoint, so they share at least one element. Then

$$ph_i = qh_j \quad \text{for some } h_i, h_j \in H.$$

Multiply on the right by h_i^{-1} :

$$p = qh_jh_i^{-1} \quad \implies \quad q^{-1}p = h_jh_i^{-1}. \quad (1)$$

Since H is a group, $h_i \in H$ gives $h_i^{-1} \in H$, and closure then gives $h_jh_i^{-1} \in H$. Combining with (1),

$$q^{-1}p \in H. \quad (2)$$

By the rearrangement lemma from Lecture 4, if $x \in H$ then $xH = H$. Applying this to (2),

$$(q^{-1}p)H = H. \quad (3)$$

Multiplying (3) on the left by q ,

$$q(q^{-1}p)H = qH \quad \implies \quad \boxed{pH = qH}$$

So if two cosets overlap at all, they are the same set. If that assumption fails, the cosets are disjoint. This is the lemma stated briefly in section 2.5 of Wu-Ki Tung.

Example. Take $H_4 = \{e, (1\ 2\ 3), (3\ 2\ 1)\}$ in D_3 , the normal subgroup from Lecture 4, and work out every left coset:

$$\begin{aligned} (1\ 2)H_4 &= \{(1\ 2), (2\ 3), (3\ 1)\} & (1\ 2\ 3)H_4 &= \{(1\ 2\ 3), (3\ 2\ 1), e\} \\ (2\ 3)H_4 &= \{(2\ 3), (3\ 1), (1\ 2)\} & (3\ 2\ 1)H_4 &= \{(3\ 2\ 1), e, (1\ 2\ 3)\} \\ (3\ 1)H_4 &= \{(3\ 1), (1\ 2), (2\ 3)\} & eH_4 &= \{e, (1\ 2\ 3), (3\ 2\ 1)\} \end{aligned}$$

Six elements, but only two distinct cosets. The first three are all $(1\ 2)H_4$ and the last three are all eH_4 . Nothing partially overlaps.

7.2 Multiplying cosets

If the cosets partition G , can we make the set of cosets into a group in its own right? The natural rule is

$$pH \circ qH = (pq)H,$$

multiply the labels and take the coset of the result.

Trying this on D_3 with $H_1 = \{e, (12)\}$, whose cosets are

$$eH_1 = \{e, (12)\}, \quad (123)H_1 = \{(123), (31)\}, \quad (321)H_1 = \{(321), (23)\},$$

the rule seems to behave. It gives

$$eH_1 \circ (123)H_1 = (123)H_1, \quad (123)H_1 \circ (321)H_1 = eH_1,$$

so eH_1 acts as the identity and $(123)H_1, (321)H_1$ are inverses of each other. Associativity checks out too.

But the rule is stated in terms of p and q , and a coset does not come with a preferred label. $(123)H_1$ and $(31)H_1$ are the same set, as are $(321)H_1$ and $(23)H_1$. Using the second labels instead,

$$(31) \circ (23) = (321) \quad \implies \quad (31)H_1 \circ (23)H_1 = (321)H_1,$$

where the first labels gave eH_1 . Same two cosets, different answer. The rule is not well defined for H_1 , so this set of cosets is not a group.

What went wrong is that H_1 is not normal. Normality is exactly the condition that makes $pH \circ qH = (pq)H$ independent of which labels you pick. We'll discuss more on this in the next lecture.

7.3 Quotient groups

Quotient group. For H a normal subgroup of G , the *quotient group* or *factor group* is the set of cosets

$$G/H = \{gH \mid g \in G\},$$

with multiplication $pH \circ qH = (pq)H$.

Example. From the section above, $H_4 = \{e, (123), (321)\}$ has just two cosets in D_3 :

$$D_3/H_4 = \{H_4, (12)H_4\} \cong C_2, \quad (12)H_4 = \{(12), (23), (31)\}.$$

H_4 is the identity, and $(12)H_4$ squares to H_4 . This is exactly the map φ from the start of Lecture 6.

Lecture 8

The Coset Product and Homomorphisms

8.1 Products of sets

Before multiplying cosets we need to say what it means to multiply sets. For $X, Y \subseteq G$,

$$X \circ Y = \{ xy \mid x \in X, y \in Y \},$$

every product of an element of X with an element of Y , collected into one set. Repeats count once.

8.2 The coset product

Cosets are sets, so this is the natural thing to try. For $H \subseteq G$ and $p, q \in G$,

$$pH \circ qH = \{ ph_iqh_j \mid h_i, h_j \in H \}.$$

Take $H_1 = \{e, (1\ 2)\}$ in D_3 , with cosets

$$eH_1 = \{e, (1\ 2)\}, \quad (1\ 2\ 3)H_1 = \{(1\ 2\ 3), (3\ 1)\}, \quad (3\ 2\ 1)H_1 = \{(3\ 2\ 1), (2\ 3)\}.$$

Then

$$eH_1 \circ (1\ 2\ 3)H_1 = \{(1\ 2\ 3), (3\ 1), (2\ 3), (3\ 2\ 1)\},$$

which has four elements. Cosets of H_1 have two. So the product of two cosets is not a coset, and there is no group here. This is the same failure as in Lecture 7, seen as a set product rather than as an ambiguity of labels.

Now take the normal subgroup $H_4 = \{e, (1\ 2\ 3), (3\ 2\ 1)\}$, with cosets

$$eH_4 = \{e, (1\ 2\ 3), (3\ 2\ 1)\}, \quad (1\ 2)H_4 = \{(1\ 2), (3\ 1), (2\ 3)\}.$$

Then

$$eH_4 \circ (1\ 2)H_4 = \{(1\ 2), (3\ 1), (2\ 3)\} = (1\ 2)H_4.$$

Nine products, but only three distinct results, and they assemble into a coset again.

Coset product. For H a normal subgroup of G ,

$$\boxed{pH \circ qH = (pq)H}$$

This is defined only for normal subgroups.

Why normality is what makes this work. Start from the set product,

$$pH \circ qH = \{ ph_iqh_j \mid h_i, h_j \in H \}.$$

Since H is normal, $Hq = qH$, so for each h_i there is some $h_k \in H$ with $h_i q = q h_k$. Then

$$p h_i q h_j = p q h_k h_j = (p q) h_\ell,$$

where $h_\ell = h_k h_j \in H$ by closure. Every element of the set product lies in $(p q)H$. Taking $h_i = h_j = e$ and letting h_j range over H recovers all of $(p q)H$, so the two sets are equal.

Without normality the step $h_i q = q h_k$ is unavailable, and the product spills outside any single coset, as it did for H_1 .

8.3 D_3/H_4 is a group

Write $D_3 = H \cup M$ with $H = H_4$ the rotations and $M = (1\ 2)H_4$ the reflections. Check the axioms directly on $G' = \{H, M\}$:

$$\begin{aligned} H \circ H &= eH \circ eH = (ee)H = H & M \circ H &= M \\ H \circ M &= eH \circ (1\ 2)H = M & M \circ M &= H \end{aligned}$$

Associativity is inherited from D_3 . The identity is H . Each element is its own inverse, $H \rightarrow H$ and $M \rightarrow M$. So G' is a group of order 2, and

$$\frac{D_3}{H_4} \cong C_2.$$

8.4 Homomorphisms

We can now give the definition promised in Lecture 6.

Homomorphism. A map $\phi : G \rightarrow G'$, where G has operation $\circ$ and G' has operation $\circ'$, that preserves group multiplication: for $g_1, g_2, g_3 \in G$,

$$g_1 \circ g_2 = g_3 \quad \implies \quad \phi(g_1) \circ' \phi(g_2) = \phi(g_3).$$

Isomorphism. A homomorphism that is one-to-one. This matches the definition from Lecture 5.

Automorphism. A one-to-one mapping from G to itself that preserves multiplication.

Natural homomorphism. If G has a normal subgroup H , there is always a homomorphism from G to the factor group:

$$G \longrightarrow \frac{G}{H}, \quad g \longmapsto gH.$$

For D_3 this recovers the picture from the start of Lecture 6. Writing $\phi : D_3 \rightarrow C_2$ and $f : D_3/H \rightarrow C_2$,

$$\phi(h) = e \text{ for } h \in H, \quad \phi(m) = a \text{ for } m \in M, \quad f(H) = e, \quad f(M) = a.$$

The first sends elements of D_3 to C_2 , the second sends cosets to C_2 . The factor group is what sits in between.

D_3 is already a group of symmetries, of the triangle. The factor group is a symmetry of that group: the split into rotations and reflections is itself respected by the group operation, since composing two reflections gives a rotation.

8.5 Conjugate elements

Conjugate elements. An element $b \in G$ is *conjugate* to $a \in G$ if there exists $p \in G$ with

$$b = pap^{-1},$$

written $a \sim b$.

In D_3 , $(1\ 2) \sim (3\ 1)$, since

$$(2\ 3) \circ (1\ 2) \circ (2\ 3)^{-1} = (3\ 1),$$

using $(2\ 3)^{-1} = (2\ 3)$ and the Cayley table from Lecture 3.

Homework

Read section 2.4 of Wu-Ki Tung on conjugacy classes. Conjugacy is a type of equivalence relation, and the classes it produces are the equivalence classes.

Supplement A

Conjugacy Classes

This material was assigned as reading rather than covered in lecture. It follows section 2.4 of Wu-Ki Tung together with the corresponding parts of Herstein.

A.1 Conjugation is an equivalence relation

Recall from Lecture 8 that $a \sim b$ means $b = pap^{-1}$ for some $p \in G$. All elements in a conjugacy relation have the same order.

Claim. Conjugation is an equivalence relation, meaning it is reflexive ($a \sim a$), symmetric ($a \sim b \Rightarrow b \sim a$) and transitive ($a \sim b$ and $b \sim c \Rightarrow a \sim c$).

Proof. For reflexivity, take $p = e$:

$$a = e^{-1}ae \quad \implies \quad a \sim a.$$

For symmetry, suppose $a \sim b$, so $b = pap^{-1}$ with $p \in G$. Rearranging,

$$a = p^{-1}b(p^{-1})^{-1},$$

and $p^{-1} \in G$, so writing $r = p^{-1}$ gives $a = rbr^{-1}$, that is $b \sim a$.

For transitivity, suppose $a \sim b$ and $b \sim c$, so

$$b = pap^{-1}, \quad c = rbr^{-1}, \quad p, r \in G.$$

Substituting,

$$c = rpap^{-1}r^{-1} = (rp)a(rp)^{-1},$$

and $rp \in G$ by closure, so $a \sim c$. □

Since conjugation is an equivalence relation, it partitions G into disjoint equivalence classes.

A.2 Conjugacy classes

Conjugacy class. The elements of a group that are conjugate to each other form a *conjugacy class*. For $a \in G$,

$$C(a) = \{x \in G \mid a \sim x\} = \{x^{-1}ax \mid x \in G\}.$$

Every element belongs to exactly one class.

The identity forms a class by itself, since $geg^{-1} = e$ for every g . Such an element is called *self-conjugate*.

In an abelian group every element is its own class:

$$gxg^{-1} = gg^{-1}x = x.$$

Both cosets and conjugacy classes partition the group, but they are different partitions. Cosets of a given subgroup all have the same size; conjugacy classes need not. In D_3 the classes are

$$\{e\}, \quad \{(123), (321)\}, \quad \{(12), (23), (31)\},$$

of sizes 1, 2 and 3.

Classes of S_n . The number of conjugacy classes in S_n is $p(n)$, the number of partitions of n . This is in Herstein.

A.3 The centralizer and the class equation

Normalizer of an element. For $a \in G$,

$$N(a) = \{x \in G \mid xa = ax\},$$

the elements of G that commute with a . This is Herstein's name for it; it is more commonly called the *centralizer* of a . It is a subgroup of G .

The size of a class is fixed by the size of this subgroup:

$$|C(a)| = \frac{|G|}{|N(a)|}.$$

Summing over the classes gives the **class equation**:

$$|G| = \sum \frac{|G|}{|N(a)|},$$

one term for each class, with a any representative of that class.

A.4 The centre

Centre. The set of all self-conjugate elements of G is called the *centre*:

$$Z(G) = \{a \in G \mid ax = xa \text{ for all } x \in G\}.$$

For an abelian group $Z(G) = G$.

If $a \in Z(G)$ then $N(a) = G$, so a contributes a class of size 1. Splitting those terms off the class equation,

$$|G| = \sum_{a \notin Z(G)} \frac{|G|}{|N(a)|} + |Z(G)|.$$

Claim. $Z(G)$ is a normal subgroup of G .

Proof. We use the subgroup test: a non-empty $H \subseteq G$ is a subgroup if $ab^{-1} \in H$ whenever $a, b \in H$.

$Z(G)$ is non-empty, since $e \in Z(G)$. Let $z_1, z_2 \in Z(G)$. First, $z_2x = xz_2$ for all x , and multiplying by z_2^{-1} on both sides gives

$$xz_2^{-1} = z_2^{-1}x \implies z_2^{-1} \in Z(G).$$

Then

$$z_1 z_2^{-1} x = z_1 (z_2^{-1} x) = z_1 (x z_2^{-1}) = (z_1 x) z_2^{-1} = (x z_1) z_2^{-1} = x z_1 z_2^{-1},$$

so $z_1 z_2^{-1} \in Z(G)$ and $Z(G)$ is a subgroup. For normality, take $z \in Z(G)$ and any $x \in G$:

$$x z x^{-1} = z x x^{-1} = z \in Z(G),$$

so $Z(G)$ is unchanged under conjugation by any element. □

This is the notion left unnamed in Lecture 6, where we noted that a normal subgroup need not consist of elements commuting individually with everything. $Z(G)$ is exactly the set of elements that do.

A.5 Conjugate subgroups

If H is a subgroup of G and $a \in G$, then

$$H' = \{ a h a^{-1} \mid h \in H \}$$

is also a subgroup of G , called a *conjugate subgroup*, written $H' \sim H$.

Invariant subgroup. A subgroup of G identical to all of its conjugate subgroups. This is the same condition as normality, in the form $g H g^{-1} = H$ from Lecture 6.

For example, with $H = \{e, (1\ 2\ 3), (3\ 2\ 1)\}$ in D_3 ,

$$(1\ 2) \{e, (1\ 2\ 3), (3\ 2\ 1)\} (1\ 2)^{-1} = \{e, (3\ 2\ 1), (1\ 2\ 3)\},$$

the same set with two elements exchanged.

Normalizer of a subgroup. For H a subgroup of G ,

$$N(H) = \{ x \in G \mid x H = H x \}.$$

Simple group. A group containing no non-trivial invariant subgroup.

Semi-simple group. A group containing no non-trivial abelian invariant subgroup.

A.6 Cauchy's and Sylow's theorems

Cauchy's theorem. If p is prime and p divides $|G|$, then G has an element of order p .

Sylow's first theorem. If p is prime and p^α divides $|G|$ while $p^{\alpha+1}$ does not, then G has a subgroup of order p^α . Such a subgroup is called a *p-Sylow subgroup*.

Example. For S_3 , $|S_3| = 6 = 2 \times 3$. Since 2 divides 6 and 4 does not, a subgroup of order 2 exists. Since 3 divides 6 and 9 does not, a subgroup of order 3 exists. Both were found in Lecture 4: the three order-2 subgroups H_1, H_2, H_3 and the rotations H_4 .

Lecture 9

Direct Products and Kernels

9.1 External direct product

Let H_1 be a group under $\cdot$ and H_2 a group under $\circ$. The **external direct product** is the set of pairs

$$G = H_1 \times H_2 = \{ (h_1, h_2) \mid h_1 \in H_1, h_2 \in H_2 \},$$

with the operation acting componentwise:

$$(h_1, h_2) \odot (h'_1, h'_2) = (h_1 \cdot h'_1, h_2 \circ h'_2).$$

Each factor keeps its own operation and the two never mix.

When H_1 and H_2 happen to carry the *same* operation, the pair can be collapsed into a single product,

$$\phi((h_1, h_2)) = h_1 \cdot h_2,$$

which is what makes the next construction possible.

9.2 Internal direct product

Here we start with one group G under $\cdot$ and two subgroups $H_1, H_2 \subseteq G$, and ask when G is built out of them. The conditions are:

- (1) $h_1 h_2 = h_2 h_1$ for all $h_1 \in H_1, h_2 \in H_2$;
- (2) every $g \in G$ can be written uniquely as $g = h_1 h_2$

When these hold we write

$$\boxed{G = H_1 \otimes H_2}$$

and call it the **internal direct product**. From here on, *direct product* means the internal one unless we say otherwise.

Note that (1) is not a claim that G is abelian: it says elements of H_1 commute with elements of H_2 , not that elements of H_1 commute among themselves.

Exercise

Show that the two conditions force

$$H_1 \cap H_2 = \{e\}.$$

Lemma. If $G = H_1 \otimes H_2$ then H_1 and H_2 are both normal subgroups of G .

Proof. A subgroup H is normal when $gag^{-1} \in H$ for every $g \in G$ and every $a \in H$. So take $a \in H_1$ and any $g \in G$.

By (2), $g = h_1 h_2$ for some $h_1 \in H_1$, $h_2 \in H_2$, so $g^{-1} = h_2^{-1} h_1^{-1}$ and

$$g a g^{-1} = h_1 (h_2 a h_2^{-1}) h_1^{-1}.$$

The bracketed part simplifies: $a \in H_1$ and $h_2 \in H_2$, so by (1) they commute, giving $h_2 a h_2^{-1} = a h_2 h_2^{-1} = a$. Therefore

$$g a g^{-1} = h_1 a h_1^{-1},$$

and all three factors on the right lie in H_1 , so the product does too by closure. Swapping the roles of H_1 and H_2 gives the same for H_2 . $\square$

Where this shows up? The Standard Model gauge group is an internal direct product,

$$G_{SM} = SU(3)_C \otimes SU(2)_L \otimes U(1)_Y.$$

A Grand Unified Theory looks for a single group, such as $SU(5)$, having an internal product structure of this shape.

Homework (Wu-Ki Tung)

Let $G = H_1 \otimes H_2$. Show that

$$\frac{G}{H_1} \cong H_2 \quad \text{and} \quad \frac{G}{H_2} \cong H_1.$$

Solution. By the lemma H_1 and H_2 are normal subgroups, so both quotients are groups under the coset product from Lecture 8. Work out the cosets of H_1 , writing each $g \in G$ as $h_1 h_2$:

$$\begin{aligned} \frac{G}{H_1} &= \{ g H_1 : g \in G \} \\ &= \{ h_1 h_2 H_1 : h_1 \in H_1, h_2 \in H_2 \} \\ &= \{ (h_1 H_1)(h_2 H_1) : h_1 \in H_1, h_2 \in H_2 \} \\ &= \{ (e H_1)(h_2 H_1) : h_2 \in H_2 \} \\ &= \{ h_2 H_1 \}, \end{aligned}$$

the fourth line because $h_1 H_1 = H_1$ for $h_1 \in H_1$. So the cosets of H_1 generated by elements of H_2 are the only elements of the factor group.

That suggests the map

$$T : H_2 \longrightarrow \frac{G}{H_1}, \quad h_2 \longmapsto h_2 H_1.$$

It is onto by the calculation above. It is one-to-one because $h_2 H_1 = h'_2 H_1$ gives $h_2'^{-1} h_2 \in H_1$, and this element also lies in H_2 , so it lies in $H_1 \cap H_2 = \{e\}$ from the exercise, forcing $h_2 = h'_2$. Finally it preserves multiplication:

$$T(h_2 h'_2) = (h_2 h'_2) H_1 = (h_2 H_1)(h'_2 H_1) = T(h_2) T(h'_2).$$

So T is an isomorphism and $G/H_1 \cong H_2$. The same argument with the roles swapped gives $G/H_2 \cong H_1$.

The converse fails. If H is a normal subgroup of G and $H' = G/H$, it does not follow that $G = H \otimes H'$. Take D_3 with $H_4 = \{e, (1\ 2\ 3), (3\ 2\ 1)\}$. From Lecture 7, $D_3/H_4 \cong C_2$, and C_2 is isomorphic to any of the order-2 subgroups H_1, H_2, H_3 . But D_3 is not $H_4 \otimes H_1$: condition (1) fails, since

$$(1\ 2) \circ (1\ 2\ 3) = (2\ 3) \quad \text{while} \quad (1\ 2\ 3) \circ (1\ 2) = (3\ 1),$$

Exercise

Find a non-abelian group with at least two normal subgroups.

Answer. The quaternion group,

$$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}, \quad i^2 = j^2 = k^2 = -1, \quad ij = k, \quad ji = -k.$$

It is non-abelian, since $ij \neq ji$.

9.3 The kernel of a homomorphism

Let $\phi : G \rightarrow G'$ be a homomorphism, and let e' be the identity of G' .

Kernel. The *kernel* of ϕ is the set of elements of G that map to the identity of G' :

$$K = \{g \in G \mid \phi(g) = e'\}.$$

Two facts:

- (i) K is a normal subgroup of G ;
- (ii) $\frac{G}{K} \cong G'$.

The second is the useful one in practice: to show two groups are isomorphic, find a homomorphism onto the second and identify its kernel. Refer to Wu-Ki Tung for the proofs and further details.

9.4 Normal subgroups and conjugacy classes

A subgroup $N \subseteq G$ is normal if and only if N is a union of complete conjugacy classes of G .

The classes of D_3 from Supplement A were $\{e\}$, $\{(1\ 2\ 3), (3\ 2\ 1)\}$ and $\{(1\ 2), (2\ 3), (3\ 1)\}$. The normal subgroup $H_4 = \{e, (1\ 2\ 3), (3\ 2\ 1)\}$ is the first two classes together, while $H_1 = \{e, (1\ 2)\}$ takes only part of the third and is not normal.

This can be used to quickly check if a subgroup is normal from its conjugacy classes and helps a lot while studying groups of higher order.

Homework

Work out the conjugacy classes of Q_8 , then use them to find all its normal subgroups. Remember that a union of classes also has to be closed under multiplication to be a subgroup, and that Lagrange restricts the possible orders.

Lecture 10

Representations

10.1 Representations

Representation. A *representation* of dimension n , written $[n]$, of a group G is a homomorphism

$$D : G \longrightarrow GL(n, \mathbb{C}),$$

where $GL(n, \mathbb{C})$ is the group of non-singular $n \times n$ matrices with complex entries. Being a homomorphism, D preserves the group structure:

$$g \longmapsto D(g), \quad D(g_1 g_2) = D(g_1) D(g_2),$$

the product on the right being ordinary matrix multiplication. The matrices must be non-singular, since $D(g^{-1}) = D(g)^{-1}$ needs $D(g)$ to be invertible.

A representation replaces abstract group elements by matrices, which we can actually compute with.

Faithful representation. A representation is *faithful* if the homomorphism is an isomorphism. By Lecture 9 this happens exactly when

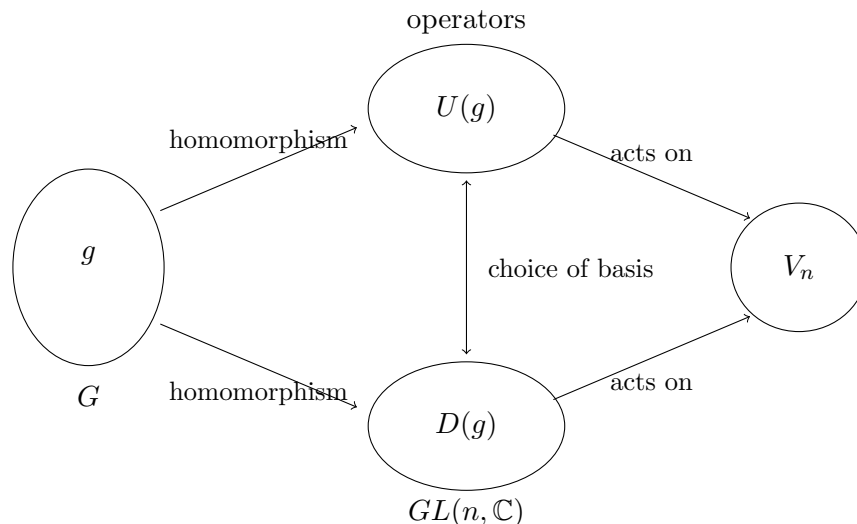
$$\ker D = \{e\},$$

so the only element sent to the identity matrix is the identity of G . A representation that is not faithful is called *degenerate*. The extreme case is the *trivial representation*, sending every element of G to the number 1, which is a valid $[1]$ representation and tells us nothing about G .

Strictly, the matrices come one step later. A group element first acts as an operator $U(g)$ on a vector space V_n , with

$$U(g_1) U(g_2) = U(g_1 g_2),$$

and the dimension of the representation is the dimension of V_n . Choosing a basis for V_n turns each operator into a matrix $D(g)$, which then acts on V_n directly. Different bases give different matrices for the same operator.



Both arrows out of G are homomorphisms. The vertical link is not: it is a choice, and matrices related by a change of basis describe the same operator.

10.2 Group actions

A representation is one instance of a more general idea: a group acting on a set.

Group action. Given a group G and a set X , an *action* is a map

$$\circ : G \times X \longrightarrow X, \quad (g, x) \longmapsto g \circ x,$$

satisfying

- (i) $e \circ x = x$ for all $x \in X$;
- (ii) $(g \cdot h) \circ x = g \circ (h \circ x)$ for all $g, h \in G$ and $x \in X$.

Condition (ii) says that multiplying in the group first and then acting is the same as acting twice in succession.

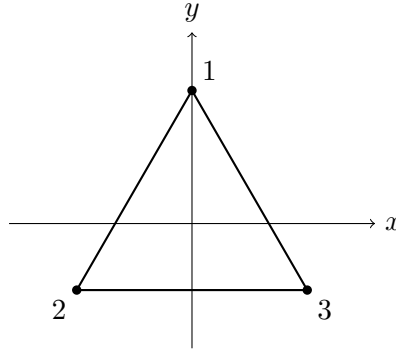
A representation is the case where X is a vector space and each g acts linearly on it.

10.3 A two-dimensional representation of D_3

The triangle lives in a plane, so its symmetries can be written as 2×2 matrices acting on that plane. Put the triangle at the origin with vertex 1 on the vertical axis, and write

$$\hat{e}_x = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \hat{e}_y = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

for the two basis vectors of the plane. Note that we don't need to define a norm here.



Write γ for the rotation by 120° and s for the reflection in the vertical axis. In cycle notation $\gamma = (1\ 2\ 3)$ and $s = (2\ 3)$, since the reflection fixes vertex 1 and swaps the other two. Every element of D_3 is a product of these two.

A rotation by θ has the matrix

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix},$$

which gives γ and γ^2 at $\theta = 120^\circ$ and 240° . For s , the reflection sends $\hat{e}_x \mapsto -\hat{e}_x$ and fixes $\hat{e}_y$. Writing s as $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, those two conditions read

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \hat{e}_x = \begin{pmatrix} a \\ c \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \hat{e}_y = \begin{pmatrix} b \\ d \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

The remaining two elements are the products γs and $\gamma^2 s$.

Rotations:

D_3	e	γ	γ^2
cycle	e	$(1\ 2\ 3)$	$(3\ 2\ 1)$
$D^{(2)}$	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$	$\begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$

Reflections:

D_3	s	γs	$\gamma^2 s$
cycle	$(2\ 3)$	$(1\ 2)$	$(3\ 1)$
$D^{(2)}$	$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$	$\begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$

The cycle labels are consistent with the Cayley table from Lecture 3. For instance γs has s acting first, and the table gives $(1\ 2\ 3) \circ (2\ 3) = (1\ 2)$, which is the entry listed.

Tracking $\hat{e}_y$ makes the geometry visible. Under γ and γ^2 ,

$$\hat{e}_y \mapsto \begin{pmatrix} -\frac{\sqrt{3}}{2} \\ -\frac{1}{2} \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} \frac{\sqrt{3}}{2} \\ -\frac{1}{2} \end{pmatrix},$$

the two points below the axis at 120° from where it started, which is where vertex 1 ends up.

This representation is faithful, and so is the [3] one you would get by writing each permutation of the three vertices as a 3×3 matrix. A group has many permutations!

10.4 Equivalent representations

Equivalence. Two $[n]$ representations $D^{(1)}$ and $D^{(2)}$ of a group G are *equivalent* if all their matrices are related by the same similarity transformation:

$$D^{(1)}(g) = S D^{(2)}(g) S^{-1} \quad \text{for all } g \in G,$$

with $S \in GL(n, \mathbb{C})$ independent of g .

From section 1, S is a change of basis, so equivalent representations are the same operators written in different bases. Equivalence is an equivalence relation in the sense of Supplement A, so representations fall into equivalence classes.

Exercise

Check that this is consistent with $D^{(2)}$ being a representation. Given $D^{(2)}(gg') = D^{(2)}(g)D^{(2)}(g')$, show that

$$D^{(1)}(gg') = D^{(1)}(g) D^{(1)}(g').$$

10.5 Reducible representations

Suppose a representation can be written in the block form

$$D(g) = \begin{pmatrix} A(g) & C(g) \\ 0 & B(g) \end{pmatrix},$$

with $A(g)$ of size $n \times n$ and $B(g)$ of size $m \times m$, and the same block sizes for every g . Multiplying two of these:

$$D(g_1)D(g_2) = \begin{pmatrix} A(g_1) & C(g_1) \\ 0 & B(g_1) \end{pmatrix} \begin{pmatrix} A(g_2) & C(g_2) \\ 0 & B(g_2) \end{pmatrix} = \begin{pmatrix} A(g_1)A(g_2) & A(g_1)C(g_2) + C(g_1)B(g_2) \\ 0 & B(g_1)B(g_2) \end{pmatrix}.$$

Reading off the diagonal blocks and using $D(g_1g_2) = D(g_1)D(g_2)$,

$$A(g_1g_2) = A(g_1)A(g_2), \quad B(g_1g_2) = B(g_1)B(g_2).$$

So $\{A(g)\}$ and $\{B(g)\}$ are themselves representations, of dimensions $[n]$ and $[m]$. A representation of this form, or equivalent to one, is *reducible*. The off-diagonal block C does not close on itself, so it gives nothing.

Completely reducible. If $C(g) = 0$ for every g , the representation is *completely reducible* and we write

$$D(g) = A(g) \oplus B(g).$$

The two blocks then act independently.

10.6 Characters

Character. The *character* of a representation is the set of traces of its matrices:

$$\chi = \{ \chi(g) \}_{g \in G}, \quad \chi(g) = \text{Tr } D(g) = \sum_i D(g)_{ii}.$$

The number does not depend on the basis which makes it pretty useful.

Characters are the same for equivalent representations. The trace is cyclic:

$$\text{Tr}(ABC) = \sum_{ijk} A_{ij} B_{jk} C_{ki} = \sum_{ijk} B_{jk} C_{ki} A_{ij} = \text{Tr}(BCA),$$

since the entries are numbers and can be reordered freely. Applying this to $D(g) = SD'(g)S^{-1}$,

$$\text{Tr}(SD'(g)S^{-1}) = \text{Tr}(D'(g)S^{-1}S) = \text{Tr } D'(g).$$

For a finite group the converse also holds: if two representations have the same character they are necessarily equivalent. This is one of the main results of representation theory.

Lecture 11

Maschke's Theorem and Vector Spaces

11.1 Similarity transformations

Exercise

Let D be an $[n]$ representation of G and $S \in GL(n, \mathbb{C})$. Show that $D' = SDS^{-1}$ is also a representation of G .

Solution. This is the exercise left at the end of Lecture 10. All that needs checking is that D' preserves multiplication:

$$D'(g_1 g_2) = S D(g_1 g_2) S^{-1} = S D(g_1) D(g_2) S^{-1},$$

using that D is a representation. Insert $S^{-1}S = \mathbb{I}$ between the two factors:

$$= SD(g_1) \underbrace{S^{-1}S}_{\mathbb{I}} D(g_2) S^{-1} = (SD(g_1)S^{-1})(SD(g_2)S^{-1}) = D'(g_1) D'(g_2).$$

So group multiplication is preserved and D' is a representation, equivalent to D by definition.

Exercise

With $D' = SDS^{-1}$ as above, show that the characters agree:

$$\chi(D(g)) = \chi(D'(g)).$$

Solution. Using the cyclic property of the trace from Lecture 10,

$$\chi(D'(g)) = \text{Tr}(SD(g)S^{-1}) = \text{Tr}(D(g)S^{-1}S) = \text{Tr} D(g) = \chi(D(g)).$$

Exercise

Show that conjugate elements have the same character: if $a \sim b$ in G , then

$$\chi(D(a)) = \chi(D(b)).$$

Solution. From Lecture 8, $a \sim b$ means $b = pap^{-1}$ for some $p \in G$. Applying D ,

$$D(b) = D(p) D(a) D(p)^{-1},$$

which is a similarity transformation with $S = D(p)$. By the previous exercise the traces agree.

So the character does not depend on the individual element, only on which conjugacy class it sits in. A character table therefore needs one column per class, not one per element.

11.2 Maschke's theorem

Recall the block form from Lecture 10,

$$D(g) = \begin{pmatrix} A(g) & C(g) \\ 0 & B(g) \end{pmatrix},$$

with blocks of sizes n and m adding to the dimension of D . When $C(g) = 0$ for every g the representation is completely reducible and we write $D = A \oplus B$.

Maschke's theorem. Every reducible representation of a finite group is completely reducible. The same holds for compact groups.

So for the groups we care about, the block form can always be brought to block-diagonal by a change of basis, and C can be assumed to vanish. This is what makes representations that cannot be reduced any further the natural building blocks: everything else is a direct sum of them.

11.3 Vector spaces

We met vector spaces briefly in Lecture 1 and can now set them out properly, since representations act on them.

Vector space. A set of vectors V together with a field of scalars F , which for us is $\mathbb{R}$ or $\mathbb{C}$. There are two operations: addition of two vectors, and multiplication of a vector by a scalar. Write $u, v, w \in V$ and $a, b \in F$.

Under addition, V is an abelian group:

- closure, $u + v \in V$;
- associativity, $(u + v) + w = u + (v + w)$;
- an identity 0 , called the *null vector*, with $u + 0 = u$;
- an inverse $-u$ for every u , with $u + (-u) = 0$;
- commutativity, $u + v = v + u$.

Scalar multiplication satisfies:

- closure, $au \in V$;
- distributivity over vector addition, $a(u + v) = au + av$;
- distributivity over scalar addition, $(a + b)u = au + bu$;
- associativity, $a(bu) = (ab)u$;
- the identity of F acts trivially, $1u = u$.

The symbols are doing double duty here. In $(a + b)u = au + bu$ the first $+$ adds two scalars in F and the second adds two vectors in V ; in $a(bu) = (ab)u$ the bracket on the right is a product of two scalars. Which is meant is always clear from whether the objects either side are scalars or vectors.

11.4 Scalar products

Scalar product. A map

$$(\cdot, \cdot) : V \times V \longrightarrow F$$

assigning a number to each ordered pair $u, v \in V$, with the properties

- **Hermiticity:** $(u, v) = (v, u)^*$;
- **Linearity:** $(w, au + bv) = a(w, u) + b(w, v)$;
- **Positivity:** $(u, u) \geq 0$, with equality *iff* $u = \text{null vector}$.

Hermiticity forces $(u, u) = (u, u)^*$, so (u, u) is real and positivity makes sense. Over $\mathbb{R}$ the conjugation does nothing and hermiticity is just symmetry.

The familiar dot product of vectors in three dimensions is the standard example. In quantum mechanics the overlap integral

$$(\psi, \varphi) = \int \psi^*(x) \varphi(x) d^3x$$

is a scalar product on a space of wavefunctions.

Given a scalar product, the *norm* of u is

$$\|u\| = (u, u)^{1/2},$$

which is real by the remark above. A vector is *normalized* if $\|u\| = 1$, and two vectors are *orthogonal* if $(u, v) = 0$.

Lecture 12

Unitary Representations and Schur's Lemma

12.1 Representations of a quotient group

Exercise

Let H be a normal subgroup of G and let D be an $[n]$ representation of the quotient group,

$$D : \frac{G}{H} \longrightarrow GL(n, \mathbb{C}).$$

Show that D also gives a representation of G .

Solution. From Lecture 8 there is a natural homomorphism

$$\phi : G \longrightarrow \frac{G}{H}, \quad \phi(g) = gH.$$

Define a map on G by following one with the other:

$$D(g) := D(gH).$$

Then for $g_1, g_2 \in G$,

$$D(g_1 g_2) = D((g_1 g_2)H) = D((g_1 H)(g_2 H)) = D(g_1 H) D(g_2 H) = D(g_1) D(g_2),$$

using the coset product in the middle step and the fact that D is a representation of G/H in the next. So group multiplication is preserved.

12.2 Invariant subspaces

Take V_{n+m} with basis $\{e_1, \dots, e_n, e_{n+1}, \dots, e_{n+m}\}$ and split it into the first n vectors and the last m . The block form from Lecture 10 is a statement about this split. Acting on a vector whose last m components vanish,

$$D(g) \begin{pmatrix} v_n \\ 0_m \end{pmatrix} = \begin{pmatrix} A(g) & C(g) \\ 0 & B(g) \end{pmatrix} \begin{pmatrix} v_n \\ 0_m \end{pmatrix} = \begin{pmatrix} A(g) v_n \\ 0_m \end{pmatrix},$$

the last m components stay zero. So the subspace spanned by $e_1, \dots, e_n$ is mapped into itself by every $D(g)$. A subspace with this property is called *invariant*, and the zero block below A is exactly the condition for it.

Irreducible representation: A representation whose only invariant subspaces are $\{0\}$ and the whole of V . Equivalently, no single change of basis brings all the matrices $D(g)$ to the block form above at once. These are the building blocks Maschke's theorem points to.

Exercise

Write down a 10-dimensional representation of D_3 . Hint: you already have a [1] and a [2], and direct sums are representations.

12.3 Orthonormal bases

Orthonormal basis: A basis with

$$(e_i, e_j) = \delta_{ij},$$

so each vector is normalized and distinct ones are orthogonal.

Any basis can be turned into one by **Gram-Schmidt**. Starting from $\{f_1, f_2, \dots\}$, normalize the first,

$$e_1 = \frac{f_1}{\|f_1\|},$$

then subtract from f_2 its component along e_1 and normalize what is left,

$$e_2 = \frac{f_2 - (e_1, f_2) e_1}{\|f_2 - (e_1, f_2) e_1\|}.$$

At each later step subtract off the components along all the e 's already built, then normalize.

12.4 Unitary representations

Unitary operator: An operator T on V is *unitary* if it preserves the scalar product:

$$(Tu, Tv) = (u, v) \quad \text{for all } u, v \in V.$$

Replacing v by $T^{-1}v$ gives the equivalent form

$$(Tu, v) = (u, T^{-1}v).$$

Now fix an orthonormal basis. The matrix of T is read off from

$$Te_j = \sum_i D_{ij} e_i, \quad \text{so} \quad D_{ij} = (e_i, Te_j).$$

Take the complex conjugate and use hermiticity, then the form above:

$$D_{ij}^* = (e_i, Te_j)^* = (Te_j, e_i) = (e_j, T^{-1}e_i) = (D^{-1})_{ji}.$$

So $D^\dagger = D^{-1}$: in an orthonormal basis a unitary operator has a unitary matrix.

Unitary representation: A representation in which every matrix $D(g)$ is unitary. Every representation of a finite group is equivalent to a unitary one, so we may assume unitarity whenever it is convenient.

Consequence for characters: If D is unitary then

$$\chi(g^{-1}) = \text{Tr } D(g)^{-1} = \text{Tr } D(g)^\dagger = (\text{Tr } D(g))^* = \chi^*(g),$$

since transposing does not change the trace.

12.5 Schur's lemma

Schur's lemma 1: Let D be an irreducible representation of G . If a matrix B commutes with every matrix of the representation,

$$B D(g) = D(g) B \quad \text{for all } g \in G,$$

then B is a multiple of the identity,

$$B = \lambda \mathbb{I}.$$

Three results follow.

- (i) **Every irreducible representation of an abelian group is one-dimensional.** In an abelian group every $D(g')$ commutes with every $D(g)$, so the lemma forces $D(g') = \lambda(g')\mathbb{I}$ for each g' . A diagonal matrix of that shape has invariant subspaces unless it is 1×1 .
- (ii) **The dimensions satisfy**

$$\sum_{\mu} d_{\mu}^2 = |G|,$$

where μ labels the inequivalent irreducible representations and d_{μ} is the dimension of the μ th. For D_3 we have the trivial [1], a second [1], and the [2] from Lecture 10:

$$1 + 1 + 4 = 6 = |D_3|.$$

- (iii) **The number of inequivalent irreducible representations equals the number of conjugacy classes.** D_3 has three classes from Supplement A, matching the three representations just listed.

12.6 The character table of D_3

From Lecture 11 the character is constant on a conjugacy class, so a table needs one column per class rather than one per element. Result (iii) says the table is square. Writing $K_1 = \{e\}$, K_2 for the three reflections and K_3 for the two rotations, with class sizes in brackets:

μ	K_1	K_2	K_3
	(1)	(3)	(2)
1	1	1	1
2	1	-1	1
3	2	0	-1

Row 1 is the trivial representation. Row 3 is the [2] representation built in Lecture 10: its identity matrix has trace 2, its reflections have trace 0, and its rotations have trace $2 \cos 120^\circ = -1$. The first column is $\chi(e)$, which is the trace of the identity matrix and so gives the dimension of each representation.

Lecture 13

Orthogonality and Character Tables

13.1 The character table of C_2

Exercise

Find all the conjugacy classes of C_2 and build its character table.

Solution: Write $C_2 = \{e, a\}$ with $a^2 = e$. The group is abelian, so from Supplement A every element is its own class:

$$K_1 = \{e\}, \quad K_2 = \{a\}, \quad |K_1| = |K_2| = 1.$$

Two classes means two inequivalent irreducible representations, by result (iii) of Lecture 12. Their dimensions satisfy

$$d_1^2 + d_2^2 = |C_2| = 2,$$

so $d_1 = d_2 = 1$, which also follows from result (i) since C_2 is abelian.

One representation sends everything to 1. For the other, $\chi(a)$ must satisfy $\chi(a)^2 = \chi(a^2) = \chi(e) = 1$, and $\chi(a) = 1$ would repeat the first, so $\chi(a) = -1$:

μ	K_1	K_2
	(1)	(1)
1 ($d = 1$)	1	1
2 ($d = 1$)	1	-1

The bracketed numbers are the class sizes, and the first column is $\chi(e)$, which gives the dimension.

13.2 The great orthogonality theorem

The tables above were forced by counting. For larger groups we need the relation that the matrix elements themselves satisfy.

Theorem: For $D^{(\mu)}$ and $D^{(\nu)}$ irreducible representations of a finite group G ,

$$\sum_g D_{ir}^{(\mu)}(g) D_{sj}^{(\nu)}(g^{-1}) = \frac{|G|}{d_\mu} \delta^{\mu\nu} \delta_{ij} \delta_{rs}.$$

If the representations are unitary, $D(g^{-1}) = D(g)^\dagger$ from Lecture 12, and the same statement reads

$$\frac{d_\mu}{|G|} \sum_g [D^{(\mu)}(g)]_{mn}^* [D^{(\nu)}(g)]_{m'n'} = \delta^{\mu\nu} \delta_{mm'} \delta_{nn'}.$$

The right-hand side vanishes unless the two representations are the same one and you are looking at the same matrix entry in both. Read as a statement about vectors, each matrix entry, as a function on G , is orthogonal to every other.

13.3 Orthogonality of characters

Setting $r = i$ and $j = s$ above and summing over both indices turns the matrix elements into traces:

$$\sum_g \chi^{(\mu)}(g) \chi^{(\nu)*}(g) = |G| \delta^{\mu\nu},$$

or dividing through,

$$\boxed{\frac{1}{|G|} \sum_g \chi^{(\mu)}(g) \chi^{(\nu)*}(g) = \delta^{\mu\nu}}$$

From Lecture 11 the character is constant on a conjugacy class, so all $|K|$ elements of a class contribute the same term and the sum can be taken over classes instead:

$$\boxed{\frac{1}{|G|} \sum_K |K| \chi^{(\mu)}(K) \chi^{(\nu)*}(K) = \delta^{\mu\nu}}$$

This is the form used in practice, since it has one term per column of the character table.

Scalar product of characters: The left-hand side is a scalar product in the sense of Lecture 11. Writing

$$\langle \phi, \chi \rangle = \frac{1}{|G|} \sum_g \phi(g) \chi^*(g),$$

the statement above is

$$\langle \chi^{(\mu)}, \chi^{(\nu)} \rangle = \delta^{\mu\nu},$$

so the characters of the irreducible representations form an orthonormal set.

Decomposing a character: By Maschke's theorem from Lecture 11, any representation is a direct sum of irreducible ones, so its character is a sum of their characters:

$$\chi = \sum_{\mu} a_{\mu} \chi^{(\mu)}.$$

Orthonormality picks out the coefficients at once:

$$a_{\mu} = \langle \chi, \chi^{(\mu)} \rangle.$$

13.4 The character table of C_3

Write $C_3 = \{e, a, a^2\}$ with $a^3 = e$. Again abelian, so

$$K_1 = \{e\}, \quad K_2 = \{a\}, \quad K_3 = \{a^2\},$$

three classes of one element each, hence three one-dimensional representations with $1 + 1 + 1 = 3 = |C_3|$.

For a one-dimensional representation the matrix is a single number, so $D(g) = \chi(g)$ and the character has to mimic group multiplication:

$$\chi(g_1 g_2) = \chi(g_1) \chi(g_2).$$

Applying this to $a^3 = e$,

$$\chi(a)^3 = \chi(e) = 1,$$

so $\chi(a)$ is a cube root of unity. Writing $\omega = e^{2\pi i/3}$, the three choices $1, \omega, \omega^2$ give the three representations, and $\chi(a^2) = \chi(a)^2$ fills in the last column:

μ	K_1	K_2	K_3
	(1)	(1)	(1)
1 ($d = 1$)	1	1	1
2 ($d = 1$)	1	ω	ω^2
3 ($d = 1$)	1	ω^2	ω

Rows 2 and 3 are orthogonal because $1 + \omega + \omega^2 = 0$.

Example of a decomposition: Take a three-dimensional representation of C_3 with character

$$\chi = (\chi(e), \chi(a), \chi(a^2)) = (3, 0, 0).$$

Only the first column contributes to each scalar product, so

$$a_\mu = \frac{1}{3}(1 \cdot 3 \cdot 1 + 0 + 0) = 1 \quad \text{for } \mu = 1, 2, 3,$$

giving

$$\chi = \chi^{(1)} + \chi^{(2)} + \chi^{(3)}.$$

Each irreducible representation appears exactly once.

13.5 The character table of D_3

Now a non-abelian case, where the table is not forced by roots of unity alone. From Supplement A the classes are

$$K_1 = \{e\}, \quad K_2 = \{(123), (321)\}, \quad K_3 = \{(12), (23), (31)\},$$

of sizes 1, 2 and 3. Three classes means three representations, and

$$1 + d_2^2 + d_3^2 = 6 \implies d_2^2 + d_3^2 = 5 \implies d_2 = 1, \quad d_3 = 2,$$

taking $d_1 = 1$ for the trivial representation. So the table looks like

μ	K_1	K_2	K_3
	(1)	(2)	(3)
1 ($d = 1$)	1	1	1
2 ($d = 1$)	1	α	β
3 ($d = 2$)	2	α'	β'

with four entries to find.

Row 2: Orthogonality against row 1, weighting each class by its size,

$$1 + 2\alpha + 3\beta = 0,$$

and normalization of row 2 against itself,

$$1 + 2|\alpha|^2 + 3|\beta|^2 = 6.$$

Solving the pair gives $\alpha = 1, \beta = -1$, or $\alpha = -\frac{7}{5}, \beta = \frac{3}{5}$. The second is ruled out because row 2 is a one-dimensional representation, so its characters obey $\chi(g)^n = \chi(g^n) = 1$ for g of order n and must have modulus 1. Hence

$$\text{row 2} = (1, 1, -1).$$

Row 3: Orthogonality against rows 1 and 2 gives

$$2 + 2\alpha' + 3\beta' = 0, \quad 2 + 2\alpha' - 3\beta' = 0.$$

Adding, $4(1 + \alpha') = 0$, so $\alpha' = -1$ and then $\beta' = 0$. Check against the normalization:

$$4 + 2|\alpha'|^2 + 3|\beta'|^2 = 4 + 2 = 6. \checkmark$$

The finished table is

μ	K_1	K_2	K_3
	(1)	(2)	(3)
1 ($d = 1$)	1	1	1
2 ($d = 1$)	1	1	-1
3 ($d = 2$)	2	-1	0

Row 3 is the two-dimensional representation built in Lecture 10: its identity matrix has trace 2, its rotations have trace $2 \cos 120^\circ = -1$, and its reflections have trace 0. Nothing about that construction was used here. The whole table came out of orthogonality and counting.

Lecture 14

Schur's Second Lemma and Decomposition

14.1 Schur's second lemma

Schur's lemma 2. Let D and D' be irreducible representations of G , acting on V and V' . Suppose there is a linear map $T : V' \rightarrow V$ with

$$T D'(g) = D(g) T \quad \text{for all } g \in G.$$

Then either $T = 0$, or $V \cong V'$ and $D' \sim D$.

14.2 Orthogonality of matrix elements

The orthogonality theorem from Lecture 13, written with daggers rather than inverses and in a form symmetric between the two representations:

$$\boxed{\frac{\sqrt{d_\mu d_\nu}}{|G|} \sum_g [D^{(\mu)}(g)^\dagger]_{nm} [D^{(\nu)}(g)]_{m'n'} = \delta^{\mu\nu} \delta_{mm'} \delta_{nn'}}$$

When $\mu = \nu$ the prefactor is $d_\mu/|G|$, recovering the earlier form.

Each matrix element gives a list of $|G|$ numbers, one per group element. Collect them into a vector, labelled by the representation and by the position in the matrix:

$$|\mu, m, n\rangle = \sqrt{\frac{d_\mu}{|G|}} \left([D^{(\mu)}(g)]_{mn} \right)_{g \in G},$$

writing $\langle \cdot | \cdot \rangle$ for the scalar product from Lecture 11. For $\mu = 3$, the $[2]$ representation of D_3 , and position $(1, 1)$, the six components are the D_{11} entries listed in the table below:

$$|3, 1, 1\rangle = \sqrt{\frac{2}{6}} \left(1, -\frac{1}{2}, -\frac{1}{2}, -1, \frac{1}{2}, \frac{1}{2} \right).$$

In this notation the boxed relation is

$$\langle \mu, m, n | \nu, m', n' \rangle = \delta^{\mu\nu} \delta_{mm'} \delta_{nn'},$$

so two of these vectors are orthogonal unless they come from the same representation and the same position. Representation μ has d_μ^2 positions, and by result (ii) of Lecture 12 the d_μ^2 add up to $|G|$. So there are $|G|$ of these vectors, each with $|G|$ components, all mutually orthogonal.

Check for D_3 . The six vectors, with the entries read off the matrices from Lecture 10 and the normalization factor dropped:

μ	entry	e	γ	γ^2	s	γs	$\gamma^2 s$
1 ($d = 1$)	D_{11}	1	1	1	1	1	1
2 ($d = 1$)	D_{11}	1	1	1	-1	-1	-1
3 ($d = 2$)	D_{11}	1	$-\frac{1}{2}$	$-\frac{1}{2}$	-1	$\frac{1}{2}$	$\frac{1}{2}$
	D_{12}	0	$-\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{2}$	0	$-\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{2}$
	D_{21}	0	$\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{3}}{2}$	0	$-\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{2}$
	D_{22}	1	$-\frac{1}{2}$	$-\frac{1}{2}$	1	$-\frac{1}{2}$	$-\frac{1}{2}$

Six vectors in six dimensions. Now take each of the four entries of the [2] representation against each of the three rows above, twelve products in all.

Against $\mu = 1$, which is just the sum of the entries:

$$\begin{aligned} D_{11} : 1 - \frac{1}{2} - \frac{1}{2} - 1 + \frac{1}{2} + \frac{1}{2} &= 0 & D_{21} : 0 + \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} + 0 - \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} &= 0 \\ D_{12} : 0 - \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} + 0 - \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} &= 0 & D_{22} : 1 - \frac{1}{2} - \frac{1}{2} + 1 - \frac{1}{2} - \frac{1}{2} &= 0 \end{aligned}$$

Against $\mu = 2$, where the last three terms carry a minus sign:

$$\begin{aligned} D_{11} : 1 - \frac{1}{2} - \frac{1}{2} + 1 - \frac{1}{2} - \frac{1}{2} &= 0 & D_{21} : 0 + \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} - 0 + \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} &= 0 \\ D_{12} : 0 - \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} - 0 + \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} &= 0 & D_{22} : 1 - \frac{1}{2} - \frac{1}{2} - 1 + \frac{1}{2} + \frac{1}{2} &= 0 \end{aligned}$$

Against $\mu = 3$ itself, taking its character $(2, -1, -1, 0, 0, 0)$, which is $D_{11} + D_{22}$ rather than a single matrix element:

$$\begin{aligned} D_{11} : 2 + \frac{1}{2} + \frac{1}{2} + 0 + 0 + 0 &= 3 & D_{21} : 0 - \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} + 0 + 0 + 0 &= 0 \\ D_{12} : 0 + \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} + 0 + 0 + 0 &= 0 & D_{22} : 2 + \frac{1}{2} + \frac{1}{2} + 0 + 0 + 0 &= 3 \end{aligned}$$

The two non-zero answers are $|G|/d_\mu = 3$, and the normalization factor $\sqrt{d_\mu/|G|} = \sqrt{1/3}$ turns them into 1.

14.3 Decomposing a representation

By Maschke's theorem any representation of a finite group is a direct sum of irreducible ones. Collecting repeats,

$$D(g) = \underbrace{D^{(1)}(g) \oplus \cdots \oplus D^{(1)}(g)}_{a_1 \text{ times}} \underbrace{D^{(2)}(g) \oplus \cdots \oplus D^{(2)}(g)}_{a_2 \text{ times}} \cdots = \bigoplus_{\mu} a_{\mu} D^{(\mu)}(g),$$

a block-diagonal matrix with a_{μ} copies of the μ th irreducible representation. The trace is the sum of the traces of the blocks,

$$\chi(g) = \sum_{\mu} a_{\mu} \chi^{(\mu)}(g),$$

and by orthonormality of characters from Lecture 13,

$$a_\mu = \langle \chi, \chi^{(\mu)} \rangle = \frac{1}{|G|} \sum_g \chi(g) \chi^{(\mu)*}(g).$$

14.4 Three-dimensional representations of D_3

Exercise

How does D_3 act on a three-dimensional space? Write down all its representations of dimension 3.

Solution. Keep the triangle in the xy -plane as in Lecture 10 and add a z -axis perpendicular to it. Each symmetry acts on the plane as before and leaves z alone, so the matrices are the 2×2 ones with a 1 in the corner. For the reflection $s = (23)$,

$$D(s) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

which is $D^{(3)} \oplus D^{(1)}$. Letting z flip under reflections instead gives $D^{(3)} \oplus D^{(2)}$, with -1 in the corner for the three reflections.

The blocks have to add to dimension 3, and from Lecture 13 the available blocks are two of dimension 1 and one of dimension 2. So either one $[2]$ and one $[1]$, giving the two above, or three $[1]$ s, giving four more from the choices of $D^{(1)}$ and $D^{(2)}$. Six in all, up to equivalence.

Lecture 15

Direct Products of Representations

15.1 The direct product of two representations

Given two representations $D^{(\mu)}$ and $D^{(\nu)}$ of a group G , of dimensions d_μ and d_ν , there is a further representation of dimension $d_\mu d_\nu$ built from them, written

$$D^{(\mu)}(g) \otimes D^{(\nu)}(g).$$

Its matrices are the ones you get by letting the two act on independent labels at once.

If $D^{(\mu)}$ and $D^{(\nu)}$ are irreducible representations of two *different* groups G_1 and G_2 , the product is an irreducible representation of the product group $G_1 \times G_2$, and every irreducible representation of $G_1 \times G_2$ arises this way. But here both factors are representations of the same G , and the same g appears in both. That representation is in general reducible, and the question is what it reduces to.

This is the situation whenever two systems are combined and the symmetry acts on both together.

Characters multiply- The character of the product is the product of the characters:

$$\chi^{(\mu \times \nu)}(g) = \chi^{(\mu)}(g) \chi^{(\nu)}(g).$$

15.2 Reducing the direct product

Applying the decomposition from Lecture 14 to the product,

$$D^{(\mu)}(g) \otimes D^{(\nu)}(g) = \bigoplus_{\sigma} a_{\sigma} D^{(\sigma)}(g)$$

with the multiplicities read off from the characters,

$$a_{\sigma} = \langle \chi^{(\mu \times \nu)}, \chi^{(\sigma)} \rangle = \frac{1}{|G|} \sum_K |K| \chi^{(\mu)}(K) \chi^{(\nu)}(K) \chi^{(\sigma)*}(K).$$

This is the *Clebsch-Gordan series*, and the a_{σ} are the Clebsch-Gordan coefficients.

15.3 Example: $2 \otimes 2$ in D_3

The character table from Lecture 13, with class sizes in brackets:

μ	K_1	K_2	K_3
	(1)	(2)	(3)
1 ($d = 1$)	1	1	1
2 ($d = 1$)	1	1	-1
3 ($d = 2$)	2	-1	0

Take $\mu = \nu = 3$, the two-dimensional representation. Squaring its character entry by entry,

$$\chi^{(3 \times 3)} = (4, 1, 0),$$

a four-dimensional representation. The multiplicities are

$$\begin{aligned} a_1 &= \frac{1}{6} [1(2 \cdot 2)1 + 2(-1 \cdot -1)1 + 3(0 \cdot 0)1] = \frac{4+2}{6} = 1 \\ a_2 &= \frac{1}{6} [1(2 \cdot 2)1 + 2(-1 \cdot -1)1 + 3(0 \cdot 0)(-1)] = \frac{4+2}{6} = 1 \\ a_3 &= \frac{1}{6} [1(2 \cdot 2)2 + 2(-1 \cdot -1)(-1) + 3(0 \cdot 0)0] = \frac{8-2}{6} = 1 \end{aligned}$$

so

$$2 \otimes 2 = D^{(1)} \oplus D^{(2)} \oplus D^{(3)},$$

with dimensions $4 = 1 + 1 + 2$. Checking against the characters directly,

$$\chi^{(1)} + \chi^{(2)} + \chi^{(3)} = (1, 1, 1) + (1, 1, -1) + (2, -1, 0) = (4, 1, 0). \checkmark$$

15.4 Orthogonality and completeness

This section follows section 3.6 of Wu-Ki Tung. The orthogonality relation from Lecture 13, written over classes,

$$\frac{1}{|G|} \sum_K |K| \chi^{(\mu)*}(K) \chi^{(\nu)}(K) = \delta^{\mu\nu},$$

says the rows of the character table are orthonormal. There is a matching statement for the columns:

$$\frac{|K|}{|G|} \sum_{\mu} \chi^{(\mu)}(K) \chi^{(\mu)*}(K') = \delta_{KK'}.$$

This is *completeness*.

15.5 A test for irreducibility

Condition for irreducibility. A representation with character χ is irreducible if and only if

$$\langle \chi, \chi \rangle = 1, \quad \text{that is} \quad \sum_K |K| |\chi(K)|^2 = |G|.$$

Proof. By Maschke's theorem, $D = \bigoplus_{\mu} a_{\mu} D^{(\mu)}$, so $\chi = \sum_{\mu} a_{\mu} \chi^{(\mu)}$. Using orthonormality of the $\chi^{(\mu)}$,

$$\langle \chi, \chi \rangle = \left\langle \sum_{\mu} a_{\mu} \chi^{(\mu)}, \sum_{\nu} a_{\nu} \chi^{(\nu)} \right\rangle = \sum_{\mu\nu} a_{\mu}^* a_{\nu} \delta^{\mu\nu} = \sum_{\mu} |a_{\mu}|^2.$$

The a_{μ} are non-negative integers, so this sum equals 1 exactly when one of them is 1 and the rest are 0, which is what it means for D to be irreducible. $\square$

Check. For the two-dimensional representation of D_3 , with $\chi = (2, -1, 0)$ and class sizes 1, 2, 3,

$$1(2)^2 + 2(-1)^2 + 3(0)^2 = 4 + 2 = 6 = |D_3|. \checkmark$$

Lecture 16

Worked Example: D_4

16.1 The problem

Question

Determine $2 \otimes 2$ as a direct sum of irreducible representations, where 2 is the two-dimensional irreducible representation of D_4 .

Nothing here is new. It uses the class structure from Supplement A, the dimension and counting results from Lecture 12, the character table method from Lecture 13, and the Clebsch-Gordan series from Lecture 15.

16.2 The group

D_4 is the symmetry group of a square, of order 8. Write a for the rotation by 90° and b for a reflection:

$$D_4 = \langle a, b \rangle, \quad a^4 = b^2 = e, \quad bab^{-1} = a^{-1}.$$

The last relation lets every element be written with the a 's on the left:

$$D_4 = \{ e, a, a^2, a^3, b, ab, a^2b, a^3b \}.$$

Labelling the vertices 1, 2, 3, 4 around the square and taking b to be the reflection through the axis between the edges 12 and 34,

$$\begin{array}{ll} a = (1\,2\,3\,4) & b = (1\,2)(3\,4) \\ a^2 = (1\,3)(2\,4) & ab = (1\,3) \\ a^3 = (1\,4\,3\,2) & a^2b = (1\,4)(2\,3) \\ & a^3b = (2\,4) \end{array}$$

So b and a^2b are the reflections through the two edge axes, and ab and a^3b the reflections through the two diagonals.

16.3 Conjugacy classes

Everything follows from $bab^{-1} = a^{-1}$, or in the rearranged form $ba = a^{-1}b$, which also gives $ba^{-1} = ab$.

Rotations. Conjugating a by b ,

$$bab^{-1} = a^{-1} = a^3,$$

so $a \sim a^3$. Conjugating a^2 by b ,

$$ba^2b^{-1} = (bab^{-1})^2 = (a^{-1})^2 = a^2,$$

and a^2 commutes with every power of a , so it commutes with everything and forms a class by itself.

Reflections. Conjugating b by a ,

$$aba^{-1} = a(ba^{-1}) = a(ab) = a^2b,$$

so $b \sim a^2b$. The same computation on ab gives

$$a(ab)a^{-1} = a^2(ba^{-1}) = a^2(ab) = a^3b,$$

so $ab \sim a^3b$. No conjugation takes an edge reflection to a diagonal one.

The five classes are

$$K_1 = \{e\}, \quad K_2 = \{a^2\}, \quad K_3 = \{a, a^3\}, \quad K_4 = \{b, a^2b\}, \quad K_5 = \{ab, a^3b\},$$

of sizes 1, 1, 2, 2, 2, adding to 8 as they must.

16.4 How many representations, and of what dimension

By result (iii) of Lecture 12, five classes means five inequivalent irreducible representations. By result (ii),

$$\sum_{\mu=1}^5 d_{\mu}^2 = |D_4| = 8.$$

One of the d_{μ} is 1, from the trivial representation. Five positive squares adding to 8 leaves only

$$1 + 1 + 1 + 1 + 4 = 8,$$

since a second 4 would already give $4 + 4 + 1 + 1 + 1 = 11$. So there are four one-dimensional representations and one two-dimensional.

16.5 The one-dimensional representations

A one-dimensional representation sends each group element to a single number, so $D(g) = \chi(g)$. Applying D to $a^4 = e$ and $b^2 = e$,

$$[D(a)]^4 = 1, \quad [D(b)]^2 = 1,$$

so $D(a)$ is a fourth root of unity and $D(b) = \pm 1$. Fourth roots would allow $D(a) = \pm i$, but the relation rules this out. Applying D to $bab^{-1} = a^{-1}$,

$$D(b)D(a)D(b)^{-1} = D(a^{-1}) = D(a^3) = [D(a)]^3,$$

and since these are numbers the left side collapses to $D(a)$. Hence

$$D(a) = [D(a)]^3 \implies [D(a)]^2 = 1 \implies D(a) = \pm 1.$$

Writing $\alpha = \chi(a)$ and $\beta = \chi(b)$, the rest of the row follows from multiplicativity: $\chi(a^2) = \alpha^2 = 1$ and $\chi(ab) = \alpha\beta$. So each row has the form

$$(1, 1, \alpha, \beta, \alpha\beta),$$

and the four sign choices give the four one-dimensional representations.

16.6 The two-dimensional representation

Write its row as $(2, x, y, z, w)$, the first entry being $\chi(e) = d_\mu = 2$. Orthogonality against each of the four rows above, weighting each class by its size, gives

$$\begin{array}{ll} \mu = 1 : & 2 + x + 2y + 2z + 2w = 0 \\ \mu = 2 : & 2 + x - 2y + 2z - 2w = 0 \end{array} \quad \begin{array}{ll} \mu = 3 : & 2 + x + 2y - 2z - 2w = 0 \\ \mu = 4 : & 2 + x - 2y - 2z + 2w = 0 \end{array}$$

Adding all four, the y, z and w terms cancel in pairs:

$$8 + 4x = 0 \implies x = -2.$$

Subtracting the second equation from the first gives $4y + 4w = 0$, and subtracting the fourth from the third gives $4y - 4w = 0$, so $y = w = 0$. The first equation then reduces to $2z = 0$, so $z = 0$. The row is

$$(2, -2, 0, 0, 0),$$

and it normalizes correctly:

$$1(2)^2 + 1(-2)^2 + 2(0)^2 + 2(0)^2 + 2(0)^2 = 8 = |D_4|. \checkmark$$

This is the representation acting on the plane of the square, the analogue of the $[2]$ representation of D_3 from Lecture 10. A rotation by θ has trace $2 \cos \theta$, giving 2 for e , 0 for a and -2 for a^2 ; every reflection in a line has trace 0.

16.7 The character table

μ	K_1	K_2	K_3	K_4	K_5
	(1)	(1)	(2)	(2)	(2)
1 ($d = 1$)	1	1	1	1	1
2 ($d = 1$)	1	1	-1	1	-1
3 ($d = 1$)	1	1	1	-1	-1
4 ($d = 1$)	1	1	-1	-1	1
5 ($d = 2$)	2	-2	0	0	0

Checking the rows. Every pair of distinct rows should give zero, with each class weighted by its size:

$$\begin{array}{ll} \mu = 1, \mu = 2 : & 1 + 1 - 2 + 2 - 2 = 0 \\ \mu = 1, \mu = 3 : & 1 + 1 + 2 - 2 - 2 = 0 \\ \mu = 1, \mu = 4 : & 1 + 1 - 2 - 2 + 2 = 0 \end{array} \quad \begin{array}{ll} \mu = 2, \mu = 3 : & 1 + 1 - 2 - 2 + 2 = 0 \\ \mu = 2, \mu = 4 : & 1 + 1 + 2 - 2 - 2 = 0 \\ \mu = 3, \mu = 4 : & 1 + 1 - 2 + 2 - 2 = 0 \end{array}$$

Row 5 against any of the others gives $2 - 2 + 0 + 0 + 0 = 0$.

Checking the columns. By completeness from Lecture 15, summing a column against itself should give $|G|/|K|$:

$$K_1 : \quad 1 + 1 + 1 + 1 + 4 = 8 = \frac{8}{1}, \quad K_3 : \quad 1 + 1 + 1 + 1 + 0 = 4 = \frac{8}{2},$$

and two different columns should give zero:

$$K_1, K_2 : \quad 1 + 1 + 1 + 1 + 2(-2) = 0.$$

16.8 Reducing $2 \otimes 2$

By Lecture 15 the character of a direct product is the product of the characters, so squaring row 5 entry by entry,

$$\chi^{(5 \times 5)} = (4, 4, 0, 0, 0),$$

a four-dimensional representation. Only K_1 and K_2 contribute to any of the sums, both of size 1, and all four one-dimensional rows carry 1 in those two columns:

$$\begin{aligned} a_1 &= \frac{1}{8} [1(4)(1) + 1(4)(1)] = \frac{8}{8} = 1 & a_4 &= \frac{1}{8} [1(4)(1) + 1(4)(1)] = 1 \\ a_2 &= \frac{1}{8} [1(4)(1) + 1(4)(1)] = 1 & a_5 &= \frac{1}{8} [1(4)(2) + 1(4)(-2)] = \frac{0}{8} = 0 \\ a_3 &= \frac{1}{8} [1(4)(1) + 1(4)(1)] = 1 \end{aligned}$$

So

$$\boxed{2 \otimes 2 = 1 \oplus 1' \oplus 1'' \oplus 1'''}$$

writing $1, 1', 1'', 1'''$ for the four one-dimensional representations. The dimensions balance, $4 = 1 + 1 + 1 + 1$, and the two-dimensional representation does not appear in its own product. Checking against the characters directly,

$$(1, 1, 1, 1, 1) + (1, 1, -1, 1, -1) + (1, 1, 1, -1, -1) + (1, 1, -1, -1, 1) = (4, 4, 0, 0, 0). \checkmark$$

Products of this kind are used in particle physics: combining two systems means taking the direct product of their representations, and the physical states are read off from the pieces it decomposes into.

Homework

The quaternion group Q_8 from Lecture 9 also has order 8. Find its conjugacy classes and character table by the same route. You should find the same list of dimensions as D_4 , and in fact the same character table, even though the two groups are not isomorphic. What does that tell you about how much the character table remembers?